# IMPROVED KISSING NUMBERS IN TWENTY-EIGHT THROUGH THIRTY-ONE DIMENSIONS

ABSTRACT. We prove that the kissing numbers in 28, 29, 30 and 31 dimensions are at least 205172, 210495, 222073 and 241079. The best previously published lower bounds were 204522, 209497, 220442 and 238354, due to Takhanov and Yun. All four constructions start from the Leech lifting of Ma et al., which replaces some minimal vectors of the Leech lattice by two or three points with short components in the extra coordinates and adds points in the extra coordinates alone. Ma et al. lift sets of 496 Leech vectors. We lift sets of 528, built from cosets of a five-dimensional subcode of the binary Golay code that contains no octad. We also add points built from Leech lattice vectors of squared norm 64 and 48 together with a direction in the extra coordinates. In dimension 28 we replace the 24 points of Ma et al. in the extra coordinates by four points in a different position, which makes room for the new points. In dimensions 29 to 31 we remove all of them and put in their place a larger number of explicit points with integer coordinates, found numerically against the fixed rest of the configuration. Each configuration is checked on all pairs by an exact program, with no symmetry reduction and no floating point.

## 1. INTRODUCTION

The kissing problem in $\mathbb{R}^n$ asks how many unit spheres can touch a central unit sphere without overlapping [3, Ch. 1, §2.2]. In terms of the points of tangency, it asks for the largest set $X$ of unit vectors in $\mathbb{R}^n$ with $\langle x, y\rangle \le 1/2$ for all distinct $x, y \in X$. We write $\tau_n$ for this maximum, as in [3, Ch. 13], and call such a set $X$ a *kissing configuration*. In dimension 24 the 196560 minimal vectors of the Leech lattice attain $\tau_{24}$ [7, 9].

For $25 \le n \le 31$ the best constructions start from this configuration. Write $n = 24 + d$. We call the first 24 coordinates the *head* and the last $d$ the *tail*. Ma et al. [8] state a general construction, which they attribute in this form to Cohn, Jiao, Kumar and Torquato [1], as improved by Kallal, Kan and Wang [6]. We call its output a *Leech lift*. It removes some pairwise disjoint sets of Leech minimal vectors. It puts each removed vector back as two or three points with short tails. It then adds points with zero head. We call the points that keep a Leech minimal vector unchanged the *body*, the points built from a removed vector the *lifts*, and the added points with zero head the *poles*. Ma et al. obtained 204520, 209496, 220440 and 238350 in dimensions 28 to 31. Takhanov and Yun [10] rotate the block of lifts and poles while keeping the body fixed, which makes room for two more points in dimension 28 and one in dimension 29. In dimension 31 they rotate only the tails of the lifts, which makes room for four more points. They also gain two points in dimension 30. This gives 204522, 209497, 220442 and 238354. Kravatskiy's public repository [4] lists larger values, 204896, 209968, 221012 and 238662, which we have not checked.

**Theorem 1.1.** *We have* $\tau_{28} \ge 205172$, $\tau_{29} \ge 210495$, $\tau_{30} \ge 222073$ *and* $\tau_{31} \ge 241079$.

Table 1 compares these values with the best published lower bounds, the repository values and the known upper bounds. The gains over the best published values are 650, 998, 1631 and 2725, and over the repository values 276, 527, 1061 and 2417.

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TABLE 1. Kissing number bounds in dimensions 28 to 31. The upper bounds are those listed in [2] (accessed October 2026). The previous lower bounds are the best published ones. The repository values are those listed in [4]; they are not published in a paper and we have not checked them. The upper bounds are the floors of numerical semidefinite programming values [5] and are not interval-certified.

| dimension | previous lower bound | repository | new lower bound | upper bound |
|---|---|---|---|---|
| 28 | 204522 [10] | 204896 | **205172** | 752292 [5] |
| 29 | 209497 [10] | 209968 | **210495** | 1075991 [5] |
| 30 | 220442 [10] | 221012 | **222073** | 1537707 [5] |
| 31 | 238354 [10] | 238662 | **241079** | 2213487 [5] |

The size of a Leech lift in dimension $24 + d$ is

$$196560 + \sum_i (|T_i| - 1)\,|S_i| + |P|,$$

where the $S_i$ are the sets of removed Leech vectors, which we call *head sets*, the $T_i$ are the sets of tails attached to them, and $P$ is the set of poles (Section 2). The tails of [8] in dimensions 28 to 31 are the roots of root systems of type $D_4$, $D_5$, $E_6$ and $E_7$, and they already make $\sum_i(|T_i| - 1)$ as large as the tails allow. We keep them. The first change is in the head sets. A head set must have pairwise inner products at most $1/4$ after normalization. Ma et al. used head sets of 496 vectors; we use 528. Adding one vector to every head set adds $\sum_i(|T_i| - 1)$ points, namely 16, 26, 48 and 84 in dimensions 28 to 31. So the larger head sets alone give $32 \cdot 16 = 512$, 832, 1536 and 2688 points more than Ma et al.

The second change adds points that are neither lifts nor poles. A *layer point* is a unit vector proportional to $(v, 8w)$, where $v$ is a vector of the Leech lattice of squared norm 64 supported on four coordinates and $w$ is a unit vector in the tail. A *hole point* has the shape of a lift, with the head replaced by a Leech lattice vector of squared norm 48. Both shapes are compatible with the body because two distinct vectors of the Leech lattice are at squared distance at least 32. Compatibility with the lifts needs more. For a layer vector $v = 4\sigma$ on a 4-set $R$ of coordinates, we need heads $h$ with $\langle h, v\rangle \le 16$ on the tail sets near the direction $w$. Lemma 4.4 gives such head sets: those on an octad through $R$ whose matching has exactly one point of $R$ in each pair. In dimension 28 we replace the 24 poles of Ma et al. by four poles at deep-hole directions of the tail root lattice, which makes room for 160 layer points.

The third change, in dimensions 29 to 31, removes all poles and puts in their place *free points*: explicit vectors with integer coordinates, found numerically against the fixed rest of the configuration and checked exactly against every other point (Section 6.3). The rest of the configuration is a structured *frame*: the body and the lifts, with 144 layer points in dimension 29 and 72 hole points in dimension 30. There are 63, 97 and 167 free points. Table 2 gives the parts of each configuration.

The paper is organized as follows. Section 2 states the Leech lift and proves its pair conditions. Section 3 describes the tails and the poles of [8]. Section 4 constructs head sets of 528 vectors. Section 5 says which head sets each dimension uses. Section 6 defines layer, hole and free points and computes the inner products of layer points with the other points. Sections 7 to 10 give the four configurations. Section 11 describes the exact checks.

Some steps of the proofs are finite facts about explicit sets that we checked by computer rather than by hand. Examples are the existence of the 512-point parts of the head sets,

TABLE 2. The parts of the four configurations.

| dimension | body | lifts | poles | layer | holes | free | total |
|---|---|---|---|---|---|---|---|
| 28 | 192336 | 12672 | 4 | 160 | 0 | 0 | 205172 |
| 29 | 189168 | 21120 | 0 | 144 | 0 | 63 | 210495 |
| 30 | 183888 | 38016 | 0 | 0 | 72 | 97 | 222073 |
| 31 | 174384 | 66528 | 0 | 0 | 0 | 167 | 241079 |

the disjointness of the head sets, the largest inner product between a given head set and a given layer vector, and, in dimensions 29 to 31, a list of explicit inequalities for the free points (Proposition 6.5). We say in each place which facts these are.

## 2. The Leech lift

Let $\Lambda \subset \mathbb{R}^{24}$ be the Leech lattice, scaled so that its minimal vectors have squared norm 32 and integer coordinates [3, Ch. 4, §11], and let $\Lambda_{\min}$ be the set of its 196560 minimal vectors. At this scale all inner products in $\Lambda$ are multiples of 8, and two distinct vectors of $\Lambda$ are at squared distance at least 32. In particular, inner products of distinct vectors of $\Lambda_{\min}$ lie in $\{0, \pm 8, \pm 16, -32\}$. The vectors of $\Lambda$ of squared norm 48 and 64 are those of type 3 and type 4 in the terminology of [3, Ch. 10].

Throughout we test compatibility exactly. For nonzero $x, y$ with entries in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ the angle between them is at least $60^\circ$ if and only if

$$\langle x, y\rangle \le 0 \quad \text{or} \quad 4\langle x, y\rangle^2 \le |x|^2|y|^2. \tag{1}$$

We call such a pair *compatible*. If every pair from a finite set of nonzero vectors is compatible, then the vectors normalized to unit length form a kissing configuration, and distinct vectors give distinct points because (1) excludes cosine 1.

A *head set* is a subset $S \subseteq \Lambda_{\min}$ with $\langle u, u'\rangle \le 8$ for all distinct $u, u' \in S$, that is, with pairwise cosines at most $1/4$.

Write $n = 24 + d$ and $\alpha = \sqrt{2/3}$, $\beta = \sqrt{1/3}$, so $\alpha^2 + \beta^2 = 1$. For $x \in \Lambda_{\min}$ let $\hat{x} = x/\sqrt{32}$, a unit vector. Throughout the paper, $b$ denotes a vector of $\Lambda_{\min}$ that stays in the body, $h$ a vector of a head set (a *head*), $t$ a tail and $p$ a pole. The data of a Leech lift are:

- pairwise disjoint head sets $S_1, \dots, S_k$;
- sets $T_1, \dots, T_k$ of unit vectors in $\mathbb{R}^d$, the *tail sets*, with $\langle t, t'\rangle \le -\frac{1}{2}$ for distinct $t, t'$ in one tail set and $\langle t, t'\rangle \le \frac{1}{2}$ for $t, t'$ in different tail sets;
- a set $P$ of unit vectors in $\mathbb{R}^d$, the *poles*, with $\langle p, p'\rangle \le \frac{1}{2}$ for distinct poles and $\langle t, p\rangle \le \frac{\sqrt{3}}{2}$ for every tail $t \in \bigcup T_i$ and pole $p$.

The configuration is $X = B \cup L \cup Z$ with

$$B = \{(\hat{b}, 0) : b \in \Lambda_{\min} \setminus \textstyle\bigcup_i S_i\} \qquad \text{(body)},$$

$$L = \{(\alpha\hat{h}, \beta t) : h \in S_i,\ t \in T_i,\ 1 \le i \le k\} \qquad \text{(lifts)},$$

$$Z = \{(0, p) : p \in P\} \qquad \text{(poles)}.$$

This is the lifting construction of [8], written with our normalization, together with the poles that [8] add after it.

**Proposition 2.1.** *$X$ is a kissing configuration in $\mathbb{R}^{24+d}$, and*

$$|X| = 196560 + \sum_{i=1}^{k} (|T_i| - 1)\,|S_i| + |P|. \tag{2}$$

*Proof.* All points of $X$ are unit vectors, so we must show that every inner product of two distinct points is at most $\frac{1}{2}$. There are six kinds of pairs. The only one that needs the head set and tail set conditions is a pair of two lifts.

*Body, body.* The inner product is $\langle b, b' \rangle / 32 \le 16/32 = \frac{1}{2}$.

*Body, lift.* The inner product is $\alpha \langle b, h \rangle / 32$. Here $b \ne h$, because the body and the head sets are disjoint, so $\langle b, h \rangle \le 16$ and the value is at most $\alpha/2 < \frac{1}{2}$.

*Lift, lift.* Take $(\alpha\hat{h}, \beta t)$ with $h \in S_i$, $t \in T_i$ and $(\alpha\hat{h}', \beta t')$ with $h' \in S_j$, $t' \in T_j$. The inner product is

$$\tfrac{2}{3} \cdot \tfrac{\langle h, h' \rangle}{32} + \tfrac{1}{3} \langle t, t' \rangle.$$

There are three cases.

- $i = j$ and $h = h'$. Then $t \ne t'$ lie in one tail set, so $\langle t, t' \rangle \le -\frac{1}{2}$. The value is at most $\frac{2}{3} - \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{2}$.
- $i = j$ and $h \ne h'$. Then $\langle h, h' \rangle \le 8$ because $S_i$ is a head set, and $\langle t, t' \rangle \le 1$. The value is at most $\frac{2}{3} \cdot \frac{1}{4} + \frac{1}{3} = \frac{1}{2}$.
- $i \ne j$. Then $h \ne h'$ because $S_i \cap S_j = \emptyset$, so $\langle h, h' \rangle \le 16$, and $\langle t, t' \rangle \le \frac{1}{2}$ because the tails lie in different tail sets. The value is at most $\frac{2}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{2}$.

*Lift, pole.* The inner product is $\beta \langle t, p \rangle \le \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2} = \frac{1}{2}$.

*Pole, pole* and *body, pole.* The inner products are $\langle p, p' \rangle \le \frac{1}{2}$ and 0.

So all cosines are at most $\frac{1}{2}$, which also shows that the points are distinct. The count is $(196560 - \sum |S_i|) + \sum |S_i||T_i| + |P|$. □

The proof never uses which head set is attached to which tail set. So any assignment of the head sets to the tail sets gives a kissing configuration. Sections 7 to 10 use this freedom.

**Lemma 2.2.** *(i) Every tail set has at most 3 tails. (ii) If the tails form a set $\mathcal{T}$, then $\sum_i (|T_i| - 1) \le \lfloor 2|\mathcal{T}|/3 \rfloor \le \lfloor 2\tau_d/3 \rfloor$.*

*Proof.* (i) For unit vectors $t_1, \dots, t_m$ with pairwise inner products at most $-\frac{1}{2}$ we have $0 \le |\sum t_j|^2 \le m - m(m-1)/2$, so $m \le 3$. (ii) All tails together form a kissing configuration in $\mathbb{R}^d$, so $|\mathcal{T}| \le \tau_d$, and $|T_i| - 1 \le \frac{2}{3}|T_i|$ by (i). □

A tail set with three tails is a *triangle*: $\{a, b, -a - b\}$ with $\langle a, b \rangle = -\frac{1}{2}$. A tail set with two tails at inner product $-1$ is an *antipodal pair*. Each triangle contributes $2|S_i|$ to (2) and each pair $|S_i|$.

## 3. Tails and poles

A tail is a unit vector; we call $\sqrt{2}\,t$ the *root* of a tail $t$. The coordinate files of Section 11 use other scalings, which we state where we use them. We use the tails of [8], as given in their published configurations. We describe them by the following finite facts, which we checked in exact arithmetic on the coordinate files. In each dimension $d \in \{4, 5, 6, 7\}$:

(1) The roots are closed under the reflections they define and have pairwise inner products in $\{0, \pm 1, -2\}$. They span $\mathbb{R}^d$, and their number is 24, 40, 72, 126. So they are the roots of a root system of type $D_4$, $D_5$, $E_6$, $E_7$; no other simply laced root system of rank $d$ has this many roots.

TABLE 3. Tails, tail sets and the poles of [8]. The last column is the largest cosine between a tail and a pole.

| dimension | $d$ | tails | tail sets | poles | tail–pole |
|---|---|---|---|---|---|
| 28 | 4 | $D_4$, 24 roots | 8 triangles | $D_4$, 24 | $\sqrt{3}/2$ |
| 29 | 5 | $D_5$, 40 roots | 12 triangles, 2 pairs | $D_5$, 40 | $(2+\sqrt{2})/4 \approx 0.8536$ |
| 30 | 6 | $E_6$, 72 roots | 24 triangles | $E_6$, 72 | $\sqrt{3}/2$ |
| 31 | 7 | $E_7$, 126 roots | 42 triangles | $E_7$, 126 | $(2+\sqrt{2})/4$ |

(2) The poles of [8], scaled to squared norm 2, have the same properties as in (1), so they form a second root system of the same type.

(3) The cosine between a tail and a pole is at most $\frac{\sqrt{3}}{2}$.

Distinct roots of a simply laced root system have cosines in $\{0, \pm\frac{1}{2}, -1\}$. So the tail set conditions of Section 2 hold for any partition of the tails into triangles $\{a, b, -a-b\}$ and antipodal pairs, and the pole–pole condition holds automatically. Facts (1) and (2) identify the two root systems. The only inner products not forced by the root system structure are those of fact (3), a check of $24^2$, $40^2$, $72^2$ and $126^2$ inner products.

In every case $\sum_i(|T_i|-1)$ equals 16, 26, 48 and 84, which is $\lfloor 2|\mathcal{T}|/3 \rfloor$, the bound of Lemma 2.2(ii). So, for these tails, the only room left in (2) is in the sizes $|S_i|$ and in the poles. The configuration in dimension 28 has four poles (Section 7). In dimensions 29 to 31 we use no poles and put free points in their place (Section 6.3). In dimension 28 the four poles are not among those of [8] either. So in all four dimensions the poles of [8] in Table 3 are used only for comparison.

In dimension 29 the tails, scaled by the same factor that gives the heads squared norm 32, are the vectors $2\sqrt{2}(\pm e_i \pm e_j)$ in the coordinates $e_1, \dots, e_5$ of our coordinate files. The 14 tail sets of [8], written without the factor $2\sqrt{2}$, are

$$\{-e_1-e_2,\ e_2+e_3,\ e_1-e_3\},\quad \{-e_1-e_4,\ e_2+e_4,\ e_1-e_2\},$$
$$\{-e_1+e_5,\ -e_2-e_5,\ e_1+e_2\},\quad \{-e_1+e_4,\ -e_4+e_5,\ e_1-e_5\},$$
$$\{-e_1+e_3,\ -e_3-e_4,\ e_1+e_4\},\quad \{-e_1+e_2,\ -e_2+e_4,\ e_1-e_4\},$$
$$\{-e_2-e_3,\ e_3+e_4,\ e_2-e_4\},\quad \{-e_2-e_4,\ e_4+e_5,\ e_2-e_5\},$$
$$\{-e_2+e_5,\ e_3-e_5,\ e_2-e_3\},\quad \{-e_2+e_3,\ -e_3-e_5,\ e_2+e_5\},$$
$$\{-e_3+e_5,\ e_4-e_5,\ e_3-e_4\},\quad \{-e_3+e_4,\ -e_4-e_5,\ e_3+e_5\},$$
$$\{\pm(e_1+e_3)\},\quad \{\pm(e_1+e_5)\}.$$

In the other dimensions the coordinates are less pleasant, and the partition of the tails into triangles is part of the coordinate files that accompany the paper. The proofs use the partition only through Proposition 2.1 and Lemma 7.1, which hold for every partition into triangles, and through the computed fact (C5) of Section 9 and the exact checks of the free points.

## 4. HEAD SETS OF 528 VECTORS

4.1. **The Golay code in the shell.** Index the coordinates of $\mathbb{R}^{24}$ by $\{0, 1, \dots, 23\}$, as in our coordinate files, and identify binary words of length 24 with subsets of $\{0, \dots, 23\}$. The vectors of $\Lambda_{\min}$ have three shapes [3, Ch. 4, §11]: 1104 of shape $(\pm 4, \pm 4, 0^{22})$, 97152 of shape $(\pm 2^8, 0^{16})$ and 98304 of shape $(\mp 3, \pm 1^{23})$. For a position $i$ and a word $a$, let $v(i,a)$ be the vector with entry $-3$ at $i$ and 1 elsewhere, with the signs of the coordinates in $a$ changed. In the description

of $\Lambda$ in [3, Ch. 4, §11], a vector with odd entries lies in $\Lambda$ if and only if its coordinate sum is 4 modulo 8 and its coordinates congruent to 3 modulo 4 form a word of a fixed copy $G$ of the extended binary Golay code. For $v(i,a)$ these coordinates are exactly $a$, and the coordinate sum is $20-2|a|$ if $i \notin a$ and $28-2|a|$ if $i \in a$. This is 4 modulo 8 for every $a \in G$, because all weights in $G$ are divisible by 4. So the vectors of shape $(\mp 3, \pm 1^{23})$ are exactly the $24 \cdot 4096$ vectors $v(i,a)$ with $a \in G$, and the supports of the vectors of shape $(\pm 2^8, 0^{16})$ are the 759 octads of the same $G$. For $c \in G$ let $S_c$ change the signs of the coordinates in $c$. It maps $\Lambda_{\min}$ to itself and sends $v(i,a)$ to $v(i,a+c)$.

**Lemma 4.1** (conflict rule)**.** *Let $a, b \in G$ and $c = a + b$. Then $\langle v(i,a), v(j,b)\rangle = 16$ in exactly these cases: $i = j$ and $c$ is an octad not containing $i$; $i \neq j$ and $c = 0$; $i \neq j$ and $c$ is an octad containing $i$ and $j$. In all other cases with $v(i,a) \neq v(j,b)$ the inner product is at most 8.*

*Proof.* Let $w = \mathrm{wt}(c)$ and $\epsilon_k = -1$ if $k \in c$, $\epsilon_k = 1$ otherwise, so that $\sum_k \epsilon_k = 24 - 2w$. The two vectors agree in sign exactly off $c$. If $i = j$, the inner product is $\sum_{k \neq i} \epsilon_k + 9\epsilon_i = 24 - 2w + 8\epsilon_i$. If $i \neq j$, it is $\sum_{k \neq i,j} \epsilon_k - 3\epsilon_i - 3\epsilon_j = 24 - 2w - 4\epsilon_i - 4\epsilon_j$. The weights in $G$ are $0, 8, 12, 16, 24$, and the values follow. □

4.2. **The 512-point part.** Let $K \leq G$ be a subcode that contains no octad. Its nonzero words have weights in $\{12, 16, 24\}$. By Lemma 4.1 with $i = j$, the vectors $v(i, a + c)$ for $c \in K$ have pairwise inner products at most 8, since no nonzero $c \in K$ is an octad. So a coset of $K$ at a fixed position $i$ gives $|K|$ vectors of a head set. Every octad-free subcode has dimension at most 5: a binary $[24, 6, 12]$ code would violate the Griesmer bound, since $12 + 6 + 3 + 2 + 1 + 1 = 25 > 24$.

We use five such subcodes $K_1, \ldots, K_5$. Each has dimension 5 and weight distribution $0^1\, 12^{28}\, 16^3$. The complements of the three words of weight 16 in $K_f$ are three pairwise disjoint octads, a *trio*. Table 4 lists a basis of each subcode and its trio. These are finite computations. The subcodes $K_4$ and $K_5$ are used only in dimension 29.

Fix $K = K_f$ and an octad $O$ of its trio. For each position $i \notin O$ choose a coset $\gamma_i \in G/K$, and let

$$A = \{v(i,a) : i \notin O,\ a \in \gamma_i\},$$

a set of $16 \cdot 32 = 512$ vectors of shape $(\mp 3, \pm 1^{23})$. We use only the 16 positions outside $O$ because the 16-point part below needs the entries $\mp 3$ of $A$ to lie outside $O$ (Lemma 4.2).

When is $A$ a head set? Two vectors at the same position are fine by the previous paragraph. For two vectors at positions $i \neq j$, Lemma 4.1 says that the inner product exceeds 8 exactly when $a + b$ is 0 or an octad through $i$ and $j$. So $A$ is a head set if and only if, for all $i \neq j$ outside $O$, the coset $\gamma_i + \gamma_j$ contains neither 0 nor an octad through $i$ and $j$. In particular the $\gamma_i$ are distinct. This condition is invariant under translation: if $(\gamma_i)$ satisfies it, so does $(\gamma_i + c)$ for every $c \in G/K$. The 128 translates are pairwise disjoint, because they use different cosets at every position. We call a head set of this form a *512-point part on* $(K, O)$.

For each pair $(K_f, O)$ that we use, an exact search found such an assignment $(\gamma_i)$. For $K_1$, an exhaustive search over all 759 octads found such assignments only on the three octads of its trio, and none on 400 random 8-sets that are not octads. We did not prove this in general, and the construction does not depend on it.

4.3. **The 16-point part.**

**Lemma 4.2.** *Let $O$ be an octad and let $A$ be a set of vectors of shape $(\mp 3, \pm 1^{23})$ whose entries $\mp 3$ all lie outside $O$. Let $M$ be a perfect matching of $O$ (four disjoint pairs covering $O$), and let $F(M)$ be the 16 vectors $(\pm 4, \pm 4, 0^{22})$ supported on the pairs of $M$, with all four sign choices on each pair. Then every vector of $F(M)$ has inner product in $\{-8, 0, 8\}$ with every vector of*

*$A$, and the vectors of $F(M)$ have pairwise inner products in $\{0, -32\}$. Conversely, every set of vectors of shape $(\pm 4, \pm 4, 0^{22})$ supported in $O$ with pairwise inner products at most 8 has at most 16 elements.*

*Proof.* The vectors of $A$ have entries $\pm 1$ on $O$, so a vector $4(\pm e_p \pm e_q)$ with $p, q \in O$ meets them in $4(\pm 1 \pm 1) \in \{-8, 0, 8\}$. Two vectors of $F(M)$ on different pairs have disjoint supports, and two on the same pair have inner product 0 or $-32$.

For the converse, two such vectors whose supports share exactly one point $p$ and agree in sign at $p$ have inner product 16. So the vectors through $p$ with a given sign at $p$ must all have the same partner, and there are at most two of them. With two signs, at most four vectors pass through $p$. Each vector passes through two of the 8 points, so there are at most $8 \cdot 4/2 = 16$ vectors. □

**Theorem 4.3.** *Let $A$ be a 512-point part on $(K, O)$ and $M$ a perfect matching of $O$. Then $A \cup F(M)$ is a head set of 528 vectors.*

*Proof.* $A$ is a head set by the choice of $(\gamma_i)$, $F(M)$ is one by Lemma 4.2, and the cross products are at most 8 by the same lemma. □

The head sets of Ma et al. have 496 vectors [8]. We do not know the largest head set.

4.4. **Disjoint head sets.** Two head sets $A \cup F(M)$ and $A' \cup F(M')$ are disjoint if and only if $A \cap A' = \emptyset$ and $M$ and $M'$ have no pair in common, because the two parts have different shapes and $F(M) \cap F(M')$ consists of the vectors on common pairs.

*Matchings.* An octad has $\binom{8}{2} = 28$ pairs and each matching uses 4, so at most 7 of our head sets lie on one octad. The three octads of one trio are disjoint, so one subcode gives at most $3 \cdot 7 = 21$ head sets. Octads from different trios share 0, 2 or 4 points, and shared pairs can be used only once. For the trios of $K_1$ and $K_2$ the six octads contain 140 distinct pairs, which allows at most 35 matchings.

*512-point parts.* Two translates of one part are disjoint. For parts of one subcode on different octads, and for parts of different subcodes, whether they meet is a condition on the translation labels. For $K_f$ and $K_g$ it depends only on the images of the labels in $G/(K_f + K_g)$, because two cosets of $K_f$ and $K_g$ at a common position meet exactly when their representatives agree modulo $K_f + K_g$. For $K_1, K_2, K_3$ we have $|K_f \cap K_g| = 4$ for $f \neq g$, so each $K_f + K_g$ has dimension 8, and a computation shows that $K_f + K_g = K_1 + K_2 + K_3$. So all conditions between parts of different subcodes live in one quotient of order 16, and a small exhaustive search finds labels that satisfy all conditions.

We do not need the details of these searches for the proofs: in each dimension the head sets are listed explicitly in the coordinate files, and we checked by computer that they are pairwise disjoint (Section 11).

4.5. **Head sets and a tetrad.** A *tetrad* is a set of four coordinates. The five octads that contain a tetrad $R$ partition the other twenty coordinates into five tetrads, and these six tetrads form a *sextet* [3, Ch. 10]. The layer points of Section 6 involve the vectors $4\sigma$ supported on a tetrad $R$, with $\sigma \in \{\pm 1\}^R$. All 16 of them lie in $\Lambda$, since $4\sigma$ is the sum of two vectors of shape $(\pm 4, \pm 4, 0^{22})$ with disjoint supports. Each has squared norm 64. We need to know how they meet head sets.

**Lemma 4.4.** *Let $R$ be a tetrad contained in an octad $O$, let $A$ be a set of vectors of shape $(\mp 3, \pm 1^{23})$ whose entries $\mp 3$ lie outside $O$, and let $M$ be a perfect matching of $O$ in which every pair has exactly one point in $R$. Then $|\langle h, 4\sigma \rangle| \leq 16$ for every $h \in A \cup F(M)$ and every*

TABLE 4. The five subcodes. Each row gives a basis: two words of weight 16 and three of weight 12. The trio consists of the complements of the weight-16 words of $K_f$.

| | |
|---|---|
| $K_1$ | $\{0,1,3,5,6,7,8,10,12,13,14,15,16,18,19,20\}$, $\{2,4,5,6,8,9,11,12,13,15,16,17,18,21,22,23\}$, $\{0,1,2,3,5,8,11,13,15,17,19,22\}$, $\{0,1,2,4,6,8,9,10,13,18,20,22\}$, $\{0,2,3,4,6,10,11,12,13,14,15,21\}$ |
| trio | $O_{11} = \{0,1,3,7,10,14,19,20\}$, $O_{12} = \{2,4,9,11,17,21,22,23\}$, $O_{13} = \{5,6,8,12,13,15,16,18\}$ |
| $K_2$ | $\{0,1,3,4,6,7,10,12,14,16,17,18,19,20,21,22\}$, $\{2,3,4,5,7,8,9,11,12,13,14,15,16,17,20,23\}$, $\{0,1,2,3,4,5,7,10,12,13,15,19\}$, $\{0,1,2,3,9,11,12,13,14,17,21,22\}$, $\{0,2,3,4,5,6,9,10,17,20,22,23\}$ |
| trio | $O_{21} = \{0,1,6,10,18,19,21,22\}$, $O_{22} = \{2,5,8,9,11,13,15,23\}$, $O_{23} = \{3,4,7,12,14,16,17,20\}$ |
| $K_3$ | $\{0,2,4,5,7,8,9,10,11,13,14,18,19,20,21,22\}$, $\{1,2,3,4,6,7,10,11,12,14,15,16,17,20,21,23\}$, $\{0,1,2,3,4,5,8,9,11,12,14,15\}$, $\{0,1,2,3,4,8,13,17,20,21,22,23\}$, $\{0,1,2,5,6,7,14,15,18,21,22,23\}$ |
| trio | $O_{31} = \{0,5,8,9,13,18,19,22\}$, $O_{32} = \{1,3,6,12,15,16,17,23\}$, $O_{33} = \{2,4,7,10,11,14,20,21\}$ |
| $K_4$ | $\{0,1,2,3,4,5,6,9,10,12,13,16,17,18,19,20\}$, $\{0,1,4,7,8,10,11,12,14,15,16,17,19,21,22,23\}$, $\{5,6,7,9,10,11,12,14,15,16,18,19\}$, $\{1,2,3,6,7,8,9,10,15,16,17,21\}$, $\{1,2,4,6,7,8,11,12,13,16,18,22\}$ |
| trio | $O_{41} = \{0,1,4,10,12,16,17,19\}$, $O_{42} = \{2,3,5,6,9,13,18,20\}$, $O_{43} = \{7,8,11,14,15,21,22,23\}$ |
| $K_5$ | $\{0,1,4,5,6,9,10,11,12,15,16,17,18,19,21,22\}$, $\{0,1,2,3,5,7,8,9,10,11,13,14,15,19,20,23\}$, $\{0,1,4,7,8,9,11,12,13,14,16,17\}$, $\{2,4,6,7,9,10,11,13,16,18,19,20\}$, $\{0,2,3,4,6,10,11,12,13,14,15,21\}$ |
| trio | $O_{51} = \{0,1,5,9,10,11,15,19\}$, $O_{52} = \{4,6,12,16,17,18,21,22\}$, $O_{53} = \{2,3,7,8,13,14,20,23\}$ |

$\sigma \in \{\pm 1\}^R$. *If instead $M$ has a pair inside $R$, then $\langle h, 4\sigma\rangle = 32$ for some $h \in F(M)$ and some $\sigma$.*

*Proof.* A vector of $A$ has entries $\pm 1$ on $R$, so it meets $4\sigma$ in $4\sum_{k\in R} \pm 1$, which has absolute value at most 16. A vector $4(\pm e_p \pm e_q)$ with exactly one of $p, q$ in $R$ meets $4\sigma$ in $\pm 16$. If $p, q \in R$, choosing the signs to agree gives 32. □

We say a matching $M$ of $O \supseteq R$ is *$R$-bipartite* if every pair of $M$ has exactly one point in $R$. The $R$-bipartite pairs of $O$ form a complete bipartite graph $K_{4,4}$, which has 16 edges, so at most four pairwise pair-disjoint $R$-bipartite matchings exist on one octad. Four exist: they form a 1-factorization of $K_{4,4}$.

**Lemma 4.5** (sextet)**.** *Every tetrad $R$ lies in exactly five octads, and any two of them meet exactly in $R$.*

*Proof.* Every 5-set lies in exactly one octad, because the octads form a Steiner system $S(5, 8, 24)$ [3, Ch. 10]. So each of the 20 points outside $R$ lies in exactly one octad through $R$, and each such octad contains 4 of them. □

So $R$-bipartite matchings on different octads through $R$ never share a pair, since a shared pair would lie in the intersection $R$. At most $5{\cdot}4 = 20$ head sets can have $R$-bipartite matchings.

TABLE 5. Number of head sets on each octad of Table 4. In dimension 29 there are also 4 head sets on $O_{41}$ and 2 on $O_{51}$.

| dimension | $O_{11}$ | $O_{12}$ | $O_{13}$ | $O_{21}$ | $O_{22}$ | $O_{23}$ | $O_{31}$ | $O_{32}$ | $O_{33}$ | total |
|---|---|---|---|---|---|---|---|---|---|---|
| 28 | 4 | – | – | 4 | – | – | – | – | – | 8 |
| 29 | 4 | – | – | 4 | – | – | – | – | – | 14 |
| 30 | – | 6 | 5 | 6 | – | 7 | – | – | – | 24 |
| 31 | 6 | 7 | 6 | 4 | 3 | 3 | 4 | 5 | 4 | 42 |

## 5. The head sets in each dimension

In each dimension the configuration contains a Leech lift built from the tails of Section 3 and one head set of Theorem 4.3 on each tail set: $k = 8$, 14, 24, 42 pairwise disjoint head sets of 528 vectors. By (2) its body and lifts together have $196560 + 528 \sum_i (|T_i| - 1)$ points, that is $196560 + 528 \cdot 16$, $+528 \cdot 26$, $+528 \cdot 48$ and $+528 \cdot 84$. Proposition 2.1 applies to any assignment of disjoint head sets to tail sets, so the assignment is free for the later sections. Table 5 lists how many head sets lie on each octad. Their matchings are pairwise pair-disjoint, and their 512-point parts are pairwise disjoint by computer check.

*Dimensions 28 and 29.* All head sets lie on octads through $S = \{0, 1, 10, 19\}$, so that they are compatible with the layer points on $S$ (Sections 7 and 8).

*Dimension 30.* Twenty-four head sets are more than the 21 that one subcode can give, so we use $K_1$ and $K_2$.

*Dimension 31.* Forty-two head sets are more than the 35 that the trios of $K_1$ and $K_2$ can give, so we use all three of $K_1, K_2, K_3$. The 42 matchings use 168 distinct coordinate pairs out of the 177 pairs that lie in the nine octads.

## 6. Layer points and hole points

We now add points to a Leech lift, and sometimes change or remove the poles. In this section we define the new points and compute their inner products with the old ones. We keep the notation of Section 2: body points $(\hat{b}, 0)$, lifts $(\alpha\hat{h}, \beta t)$ with $h$ in a head set and $t$ a tail of its tail set, and poles $(0, p)$ with $p$ a unit vector. We use the following consequence of the minimum distance of $\Lambda$: if $x, y \in \Lambda$ are distinct, then

$$\langle x, y\rangle \le \tfrac{1}{2}\big(|x|^2 + |y|^2 - 32\big). \tag{3}$$

For $|x|^2 = 64$ and $y \in \Lambda_{\min}$ this gives $\langle x, y\rangle \le 32$; for $|x|^2 = 48$ and $y \in \Lambda_{\min}$ it gives $\langle x, y\rangle \le 24$; for $|x|^2 = 48$ and $|y|^2 = 64$ it gives $\langle x, y\rangle \le 40$.

6.1. **Layer points.** Let $v \in \Lambda$ with $|v|^2 = 64$ and let $w \in \mathbb{R}^d$ be a unit vector. The *layer point* with data $(v, w)$ is the unit vector

$$Y(v, w) = \Big(\frac{v}{8\sqrt{2}},\ \frac{w}{\sqrt{2}}\Big),$$

which is proportional to $(v, 8w)$. In all our layer points, $v = 4\sigma$ is supported on a tetrad $R$ as in Section 4.5. The parity of $\sigma$ is the parity of the number of its minus signs. For two sign vectors $\sigma, \sigma'$ on $R$ that differ in $j$ places, $\langle 4\sigma, 4\sigma'\rangle = 16(4 - 2j)$. So two distinct sign vectors of the same parity give $\langle v, v'\rangle \in \{0, -64\}$, and two of different parity give $\pm 32$. If $\sigma$ is supported on $R$ and $\sigma'$ on another tetrad $R'$, then $|\langle 4\sigma, 4\sigma'\rangle| \le 16\,|R \cap R'|$.

The next two lemmas contain all the inner product bounds we need for layer points. The proofs in Sections 7 and 8 only check which case applies.

**Lemma 6.1** (one layer point)**.** *For a layer point $Y = Y(v, w)$:*

(1) $\langle Y, (\hat{b}, 0)\rangle = \langle v, b\rangle/64 \le \frac{1}{2}$ *for every* $b \in \Lambda_{\min}$*;*

(2) $\langle Y, (\alpha\hat{h}, \beta t)\rangle = \dfrac{\langle h, v\rangle}{32\sqrt{6}} + \dfrac{\langle t, w\rangle}{\sqrt{6}}$*;*

(3) $\langle Y, (0, p)\rangle = \langle w, p\rangle/\sqrt{2}$.

*The value in (2) is at most $\frac{1}{2}$ in each of the following two cases:*

(L1) $\langle h, v\rangle \le 16$ *and* $\langle t, w\rangle \le \frac{1}{\sqrt{2}}$*; the value is then at most* $\frac{16}{32\sqrt{6}} + \frac{1}{\sqrt{12}} \approx 0.4928$*;*

(L2) $\langle t, w\rangle \le 0$*; the value is then at most* $\frac{1}{\sqrt{6}} \approx 0.4082$.

*Proof.* The three formulas are direct computations. The bound in (1) is (3). In (2) we always have $\langle h, v\rangle \le 32$ by (3), which gives (L2). In (L1) the two bounds are substituted directly. □

**Lemma 6.2** (two layer points)**.** *Let $Y = Y(v, w)$ and $Y' = Y(v', w')$ be distinct layer points, with $v = 4\sigma$ on a tetrad $R$ and $v' = 4\sigma'$ on a tetrad $R'$. Then*

$$\langle Y, Y'\rangle = \frac{\langle v, v'\rangle}{128} + \frac{\langle w, w'\rangle}{2},$$

*and this is at most $\frac{1}{2}$ in each of the following cases:*

(P1) $w = w'$*, $R = R'$, and $\sigma, \sigma'$ have the same parity;*

(P2) $\langle w, w'\rangle \le 0$*;*

(P3) $\langle w, w'\rangle \le \frac{1}{2}$ *and* $|\langle v, v'\rangle| \le 32$*. The second condition holds when $R = R'$ and $\sigma, \sigma'$ have different parity, and when $|R \cap R'| \le 2$.*

*Proof.* The formula is a direct computation. Since $|v|^2 = |v'|^2 = 64$, we have $|\langle v, v'\rangle| \le 64$. In (P1) the points are distinct, so $\sigma \ne \sigma'$, and the same parity gives $\langle v, v'\rangle \le 0$; the value is at most $\frac{1}{2}$. In (P2) the value is at most $\frac{64}{128} = \frac{1}{2}$. In (P3) it is at most $\frac{32}{128} + \frac{1}{4} = \frac{1}{2}$. The two sufficient conditions for $|\langle v, v'\rangle| \le 32$ were noted above. □

6.2. **Hole points.** Let $u \in \Lambda$ with $|u|^2 = 48$ and let $y \in \mathbb{R}^d$ be a unit vector. The *hole point* with data $(u, y)$ is

$$x(u, y) = \Big(\alpha\frac{u}{\sqrt{48}},\ \beta y\Big),$$

with the same proportions as a lift. We use hole points in dimension 30, where the directions $y$ are those of minuscule weights of $E_6$; they point at deep holes of the lattice spanned by the roots. The bounds for hole points are proved in Section 9.

6.3. **Free points.** In dimensions 29 to 31 we fix a *frame*: the body and the lifts, together with the layer points in dimension 29 and the hole points in dimension 30, and no poles. The pairs inside the frame are handled by the lemmas of this paper. We then add a set $E_n$ of *free points*. Each free point is the normalization of a vector $z \in \mathbb{Z}^n$ with $|z_i| \le 30000$, and $E_{29}$, $E_{30}$, $E_{31}$ have 63, 97 and 167 points. They are printed in Appendix A and listed in the accompanying coordinate files. We found them numerically: with the frame fixed, we relaxed a penalty on all cosines above $\frac{1}{2}$ that involve the free points, added one point at a time at the deepest hole of the current configuration, and relaxed again. At the end we rounded each point to an integer vector of the stated size.

The free points have no closed-form description, so the pairs that contain one cannot be settled by a uniform argument. We reduce them to a short list of explicit inequalities. Write a free point as $z = (a, q)$ with $a \in \mathbb{Z}^{24}$ and $q \in \mathbb{Z}^d$, and let

$$\mu(a) = \max_{b \in \Lambda_{\min}} \langle a, b\rangle.$$

**Lemma 6.3** (shell maximum)**.** *For $a \in \mathbb{R}^{24}$ let $|a|_{(1)} \ge |a|_{(2)}$ be the two largest of $|a_0|, \dots, |a_{23}|$. For an octad $O$ let $\nu_O(a) = 1$ if an odd number of $i \in O$ have $a_i < 0$, and $\nu_O(a) = 0$ otherwise. For $c \in G$ let $\epsilon_k^c = -1$ if $k \in c$ and $\epsilon_k^c = 1$ otherwise. Then $\mu(a) = \max(\mu_4, \mu_8, \mu_3)$, where*

$$\mu_4 = 4\big(|a|_{(1)} + |a|_{(2)}\big),$$
$$\mu_8 = \max_O \; 2\Big(\sum_{i\in O} |a_i| - 2\nu_O(a) \min_{i\in O} |a_i|\Big),$$
$$\mu_3 = \max_{c\in G} \Big(\sum_k \epsilon_k^c a_k + 4 \max_i \big(-\epsilon_i^c a_i\big)\Big),$$

*and $O$ runs over the 759 octads of $G$.*

*Proof.* We take the maximum over each of the three shapes of Section 4; $\mu_4$, $\mu_8$ and $\mu_3$ are the maxima over the shapes $(\pm 4, \pm 4, 0^{22})$, $(\pm 2^8, 0^{16})$ and $(\mp 3, \pm 1^{23})$. All 1104 vectors $(\pm 4, \pm 4, 0^{22})$ lie in $\Lambda_{\min}$, and the best one puts $\pm 4$ with the signs of $a$ on the two largest $|a_i|$; this gives $\mu_4$. A vector of shape $(\pm 2^8, 0^{16})$ in $\Lambda$ has an octad $O$ as support, and its coordinate sum is divisible by 8 [3, Ch. 4, §11], so it has an even number of minus signs; all 128 such sign choices on each octad occur, since $97152 = 759 \cdot 128$. On $O$ the best choice follows the signs of $a$ when that choice has an even number of minus signs, that is when $\nu_O(a) = 0$, and otherwise differs from it in the one place with the smallest $|a_i|$; this gives $\mu_8$. The vectors of shape $(\mp 3, \pm 1^{23})$ are the $v(i, c)$ with $c \in G$, and $\langle v(i,c), a\rangle = \sum_k \epsilon_k^c a_k - 4\epsilon_i^c a_i$; this gives $\mu_3$. □

**Lemma 6.4** (free points against the frame)**.** *Let $z = (a, q)$ be a nonzero vector in $\mathbb{R}^{24} \times \mathbb{R}^d$.*

(1) *For $b \in \Lambda_{\min}$, the pair $z, (\hat{b}, 0)$ is compatible if and only if $\langle a, b\rangle \le \sqrt{8}\,|z|$. So $z$ is compatible with every body point if $\mu(a) \le \sqrt{8}|z|$, and also if every $b \in \Lambda_{\min}$ with $\langle a, b\rangle > \sqrt{8}|z|$ lies in one of the head sets.*

(2) *Let $T_i$ be a tail set with head set $S_i$, and let $m_i(a) = \max_{h\in S_i} \langle a, h\rangle$. Then*

$$\max_{h\in S_i,\ t\in T_i} \langle z, (\alpha\hat{h}, \beta t)\rangle = \frac{\alpha\, m_i(a)}{\sqrt{32}} + \beta \max_{t\in T_i} \langle q, t\rangle.$$

*So $z$ is compatible with every lift on $T_i$ if and only if $\alpha m_i(a)/\sqrt{32} + \beta\langle q, t\rangle \le \frac{1}{2}|z|$ for each of the $|T_i|$ tails $t \in T_i$.*

*Proof.* (1) The point $(\hat{b}, 0)$ is a unit vector and $\langle z, (\hat{b}, 0)\rangle = \langle a, b\rangle/\sqrt{32}$, so (1) reads $\langle a, b\rangle \le \frac{1}{2}\sqrt{32}|z| = \sqrt{8}|z|$. The body is $\Lambda_{\min}$ minus the head sets. (2) The lifts on $T_i$ are the points $(\alpha\hat{h}, \beta t)$ for all pairs $(h, t) \in S_i \times T_i$, and the inner product is $\alpha\langle a, h\rangle/\sqrt{32} + \beta\langle q, t\rangle$, a sum of a term in $h$ and a term in $t$. The maximum over the product set is the sum of the two maxima. □

**Proposition 6.5** (finite check)**.** *For $n = 29, 30, 31$, every pair of the configuration of Section 8, 9 or 10 that contains a free point is compatible. More precisely, the following exact inequalities hold; Table 6 gives their number.*

(F1) *For each free point, $\mu(a) \le \sqrt{8}|z|$, computed with Lemma 6.3, except for the free points counted in the table; for those, every $b \in \Lambda_{\min}$ with $\langle a, b\rangle > \sqrt{8}|z|$ is a head.*

(F2) *For each free point and each tail $t$ of each tail set $T_i$, $\alpha m_i(a)/\sqrt{32} + \beta\langle q, t\rangle \le \frac{1}{2}|z|$.*

(F3) *Each free point is compatible with each layer point (dimension 29) and each hole point (dimension 30).*

(F4) *Any two free points are compatible.*

TABLE 6. The exact inequalities of Proposition 6.5. In (F1) the first number counts the free points with $\mu(a) > \sqrt{8}|z|$ and the second the vectors $b \in \Lambda_{\min}$ above the bound for them, all of which are heads. The last column is the largest cosine of a pair that contains a free point.

| $n$ | free points | (F1) | (F2) | (F3) | (F4) | largest cosine |
|---|---|---|---|---|---|---|
| 29 | 63 | 1, 2 | 2520 | 9072 | 1953 | 0.499823 |
| 30 | 97 | 19, 34 | 6984 | 6984 | 4656 | 0.499825 |
| 31 | 167 | 28, 50 | 21042 | 0 | 13861 | 0.499842 |

*Proof.* By Lemma 6.4, (F1) and (F2) cover the pairs of a free point with the body and with the lifts; (F3) and (F4) are the remaining pairs. Each inequality has the form $X \le \frac{1}{2}|x||y|$ with $X$, $|x|^2$ and $|y|^2$ in $\mathbb{Z}[\sqrt{2}, \sqrt{3}]$. We decide it as in (1): $X \le 0$, or $4X^2 \le |x|^2|y|^2$, with the signs decided exactly as in Section 11. In (F1) and (F4) all of these quantities are integers. The free points are printed in Appendix A, and the head sets, tails, layer points and hole points are the explicit sets of Sections 4 to 10, listed in the coordinate files. We carried out all these comparisons by computer. The exact all-pairs program of Section 11 decides the same pairs. □

So all pairs with a free point have cosine below 0.49985: none of them is tight.

## 7. DIMENSION 28

7.1. **The frame.** The 24 tails are the unit vectors of a root system of type $D_4$. Choose an orthonormal frame $e_1, \dots, e_4$ of $\mathbb{R}^4$ in which the tails are $(\pm e_i \pm e_j)/\sqrt{2}$. The 24 unit vectors

$$\Pi_0 = \{\pm e_i\}, \qquad \Pi_+ = \{\tfrac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1) \text{ with an even number of minus signs}\},$$
$$\Pi_- = \{\tfrac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1) \text{ with an odd number of minus signs}\}$$

are the directions of the deep holes of the $D_4$ root lattice. Each of $\Pi_0, \Pi_+, \Pi_-$ is the vertex set of a cross-polytope: two of its vectors have inner product 0 or $-1$. Vectors of different classes have inner product $\pm\frac{1}{2}$. A tail has inner product 0 or $\pm\frac{1}{\sqrt{2}}$ with each of the 24 vectors. These facts are short computations in the frame.

**Lemma 7.1.** *Let $c \in \Pi_0$. Of the eight triangles of tails, exactly two are perpendicular to $c$, and each of the other six has exactly one tail $t$ with $\langle t, c\rangle = \frac{1}{\sqrt{2}}$.*

*Proof.* For a triangle $\{a, b, -a-b\}$ the three values $\langle a, c\rangle, \langle b, c\rangle, -\langle a, c\rangle - \langle b, c\rangle$ lie in $\{0, \pm\frac{1}{\sqrt{2}}\}$ and sum to 0. So either all are 0, or they are $0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$ in some order. Twelve of the 24 tails are not perpendicular to $c$, so six triangles are of the second kind and two of the first. □

7.2. **The head sets.** Let $S = \{0, 1, 10, 19\}$. The octads $O_{11} = S \cup \{3, 7, 14, 20\}$ and $O_{21} = S \cup \{6, 18, 21, 22\}$ both contain $S$. We use four head sets on each: $A_j \cup F(M_j)$ for $j = 1, \dots, 8$, where $A_1, \dots, A_4$ are pairwise disjoint 512-point parts on $(K_1, O_{11})$ and $A_5, \dots, A_8$ on $(K_2, O_{21})$, and where $M_1, \dots, M_4$ and $M_5, \dots, M_8$ are the two 1-factorizations of the $S$-bipartite pairs of $O_{11}$

and $O_{21}$:

$$M_1 = \{0\text{-}14, 19\text{-}20, 1\text{-}3, 10\text{-}7\},\ M_2 = \{0\text{-}20, 19\text{-}14, 1\text{-}7, 10\text{-}3\},$$
$$M_3 = \{0\text{-}3, 19\text{-}7, 1\text{-}14, 10\text{-}20\},\ M_4 = \{0\text{-}7, 19\text{-}3, 1\text{-}20, 10\text{-}14\},$$
$$M_5 = \{0\text{-}21, 1\text{-}22, 10\text{-}6, 19\text{-}18\},\ M_6 = \{0\text{-}22, 1\text{-}21, 10\text{-}18, 19\text{-}6\},$$
$$M_7 = \{0\text{-}6, 1\text{-}18, 10\text{-}21, 19\text{-}22\},\ M_8 = \{0\text{-}18, 1\text{-}6, 10\text{-}22, 19\text{-}21\}.$$

The matchings are pairwise pair-disjoint by Lemma 4.5. The disjointness of the 512-point parts on different octads is a computer check. By Lemma 4.4, every head $h$ of these eight sets has $|\langle h, 4\sigma\rangle| \le 16$ for all $\sigma \in \{\pm1\}^S$.

7.3. **The configuration.** Fix two orthogonal vectors $p_1, p_2 \in \Pi_0$ and let $R_1 = \{0, 1, 6, 18\}$ and $R_2 = \{0, 3, 7, 19\}$. Both are tetrads: $R_1 \subset O_{21}$ and $R_2 \subset O_{11}$, and $|S \cap R_1| = |S \cap R_2| = 2$, $|R_1 \cap R_2| = 1$. The configuration consists of:

- the body and the lifts of the eight head sets above on the eight triangles, with the assignment described below: $192336 + 12672$ points;
- poles at the four vectors of $\Pi_0 \setminus \{\pm p_1, \pm p_2\}$;
- the *S-layer*: 64 layer points $Y(4\sigma, w)$ with $w \in \Pi_+$ and $\sigma \in \{\pm1\}^S$ even, and 64 layer points $Y(4\sigma, w)$ with $w \in \Pi_-$ and $\sigma$ odd;
- the *R-layer*: 16 layer points $Y(4\tau, \pm p_1)$ with $\tau \in \{\pm1\}^{R_1}$ even, and 16 layer points $Y(4\tau, \pm p_2)$ with $\tau \in \{\pm1\}^{R_2}$ odd.

*Assignment of head sets to triangles.* The $S$-layer is compatible with every head set, by Lemma 4.4. The $R$-layer is not, and this is what the assignment has to handle. For $j = 1, \dots, 8$ let

$$m_1(j) = \max\{\langle h, 4\tau\rangle : h \in A_j \cup F(M_j),\ \tau \in \{\pm1\}^{R_1} \text{ even}\},$$

and define $m_2(j)$ in the same way with $R_2$ and odd $\tau$. The values are

$$m_1(j) = \begin{cases} 32 & j = 7, 8, \\ 16 & \text{otherwise}, \end{cases} \qquad m_2(j) = \begin{cases} 32 & j = 3, 4, \\ 16 & \text{otherwise}. \end{cases}$$

Lemma 4.4 gives these values for the sets on the octad that contains the tetrad: $M_5, M_6$ are $R_1$-bipartite and $M_7, M_8$ have a pair inside $R_1$; $M_1, M_2$ are $R_2$-bipartite and $M_3, M_4$ have a pair inside $R_2$. The value 16 for the four sets on the other octad is a computer check. It is not automatic: a head of shape $(\mp3, \pm1^{23})$ whose entry $\mp3$ lies in the tetrad can reach 24, and the check shows that the parity condition on $\tau$ rules this out for these sets.

We put the sets $j = 7, 8$ on the two triangles perpendicular to $p_1$ and the sets $j = 3, 4$ on the two triangles perpendicular to $p_2$ (Lemma 7.1). This is possible because no triangle is perpendicular to both: the tails perpendicular to $p_1$ and $p_2$ are the four vectors $(\pm e_k \pm e_l)/\sqrt{2}$ in the remaining two coordinates, and they contain no triangle. The other four sets go on the other four triangles in any order. The result is:

(4) if a triangle is not perpendicular to $p_i$, then its head set has $m_i(j) = 16$.

**Theorem 7.2.** *This is a kissing configuration of* $192336 + 12672 + 4 + 128 + 32 = 205172$ *points in* $\mathbb{R}^{28}$.

*Proof.* Pairs among the body and the lifts are covered by Proposition 2.1. The remaining pairs involve a pole or a layer point. Pairs with a pole are easy; for pairs with a layer point we check the cases of Lemmas 6.1 and 6.2.

*Pole, lift* and *pole, pole.* A tail and a vector of $\Pi_0$ have inner product at most $\frac{1}{\sqrt{2}}$, so a lift and a pole have inner product at most $\beta/\sqrt{2} = \frac{1}{\sqrt{6}}$. Two poles have inner product 0 or $-1$. Poles have zero head, so pairs with the body are orthogonal.

*Layer, body.* Lemma 6.1(1).

*Layer, lift.* A tail $t$ has $\langle t, w\rangle \le \frac{1}{\sqrt{2}}$ for every $w$ in $\Pi_0 \cup \Pi_+ \cup \Pi_-$. For the $S$-layer, $\langle h, 4\sigma\rangle \le 16$ for every head by Lemma 4.4, so (L1) applies. For an $R$-layer point with direction $\pm p_i$ and a lift on a triangle not perpendicular to $p_i$, (4) gives $\langle h, 4\tau\rangle \le 16$, so (L1) applies again. On a triangle perpendicular to $p_i$ we have $\langle t, w\rangle = 0$, so (L2) applies.

*Layer, pole.* By Lemma 6.1(3) the value is $\langle w, p\rangle/\sqrt{2}$. An $S$-layer direction has inner product $\pm\frac{1}{2}$ with every vector of $\Pi_0$, which gives at most $\frac{1}{2\sqrt{2}}$. The $R$-layer directions $\pm p_1, \pm p_2$ are orthogonal to the four poles.

*Layer, layer.* We check Lemma 6.2. Two points of the same part ($S$-layer in $\Pi_+$, $S$-layer in $\Pi_-$, $R$-layer at $\pm p_1$, or $R$-layer at $\pm p_2$) have the same tetrad and the same parity, and their directions are equal or have inner product 0 or $-1$: case (P1) or (P2). Between $\Pi_+$ and $\Pi_-$, the directions have inner product $\pm\frac{1}{2}$, and the parities on $S$ differ: case (P2) or (P3). Between the $S$-layer and the $R$-layer, the directions have inner product $\pm\frac{1}{2}$ and $|S \cap R_i| = 2$: again (P2) or (P3). Between the two parts of the $R$-layer, the directions are orthogonal: case (P2).

So all cosines are at most $\frac{1}{2}$. □

## 8. Dimension 29

8.1. **The frame and the head sets.** The tails, scaled as in Section 3, are $2\sqrt{2}(\pm e_i \pm e_j)$ in the coordinates $e_1, \dots, e_5$ of that section, in 12 triangles and 2 antipodal pairs. Let $S = \{0, 1, 10, 19\}$ as before. By Lemma 4.5, $S$ lies in five octads; four of them are

$$O_{11} = S \cup \{3, 7, 14, 20\}, \quad O_{21} = S \cup \{6, 18, 21, 22\},$$
$$O_{41} = S \cup \{4, 12, 16, 17\}, \quad O_{51} = S \cup \{5, 9, 11, 15\},$$

which lie in the trios of $K_1$, $K_2$, $K_4$ and $K_5$. We use 4, 4, 4 and 2 head sets $A \cup F(M)$ on them, with pairwise disjoint 512-point parts on $(K_f, O)$ and $S$-bipartite matchings $M$. The matchings on one octad are pairwise pair-disjoint, and by Lemma 4.5 so are matchings on different octads. The disjointness of the 512-point parts is a computer check. By Lemma 4.4, $|\langle h, 4\sigma\rangle| \le 16$ for every head $h$ and every $\sigma \in \{\pm 1\}^S$.

8.2. **Layer points.** Let $\mathcal{V} = \{\pm e_1, \dots, \pm e_5\}$ and let $\mathcal{W}$ be the eight vectors $\pm\frac{1}{2}(\eta_1 e_1 + \eta_2 e_2 + \eta_3 e_4 + \eta_4 e_5)$, where $\eta$ runs over the rows of a $4 \times 4$ Hadamard matrix. The 144 layer points are $Y(4\sigma, w)$ with $w \in \mathcal{V}$ and $\sigma$ even on $S$ (80 points), and with $w \in \mathcal{W}$ and $\sigma$ odd (64 points). A tail has inner product in $\{0, \pm\frac{1}{\sqrt{2}}\}$ with every $w \in \mathcal{V}$ and in $\{0, \pm\frac{1}{2\sqrt{2}}, \pm\frac{1}{\sqrt{2}}\}$ with every $w \in \mathcal{W}$. Two vectors of $\mathcal{V}$, or two of $\mathcal{W}$, have inner product 0 or $-1$; a vector of $\mathcal{V}$ and one of $\mathcal{W}$ have inner product 0 or $\pm\frac{1}{2}$.

8.3. **The configuration.** The configuration consists of the body and the lifts of the 14 head sets, the 144 layer points, and a set $E_{29}$ of 63 free points. There are no poles. The tail of each of the 63 free points has cosine at least 0.95 with a half-spin direction $s/\sqrt{5}$, $s \in \{\pm 1\}^5$, of $D_5$, which points at a deep hole of the $D_5$ root lattice; for 47 of them the cosine is at least 0.98. About twenty of them are close to points proportional to $(\lambda u, s)$ with $\lambda > 0$, $u \in \Lambda$, $|u|^2 = 48$, and about a third of the squared norm in the tail; the others carry between about 0.37 and 0.97 of their squared norm in the tail.

**Theorem 8.1.** *This configuration is a kissing configuration of* $189168 + 21120 + 144 + 63 = 210495$ *points in* $\mathbb{R}^{29}$.

*Proof.* Pairs among the body and lifts are covered by Proposition 2.1, and pairs with a free point by Proposition 6.5. Pairs with a layer point go through Lemmas 6.1 and 6.2, with no poles to consider.

*Layer, body.* Lemma 6.1(1).

*Layer, lift.* Every head has $\langle h, 4\sigma\rangle \le 16$ by Lemma 4.4, and every tail has $\langle t, w\rangle \le \frac{1}{\sqrt{2}}$: case (L1).

*Layer, layer.* All layer points lie on the tetrad $S$. Two points with directions in $\mathcal{V}$, or two with directions in $\mathcal{W}$, have the same parity, and their directions are equal or have inner product 0 or $-1$: case (P1) or (P2). A point with direction in $\mathcal{V}$ and one with direction in $\mathcal{W}$ have opposite parities and directions at inner product 0 or $\pm\frac{1}{2}$: case (P2) or (P3). □

## 9. Dimension 30

The frame is a Leech lift without poles, with the 72 tails of type $E_6$ in 24 triangles and 24 head sets on the octads of Table 5, together with 72 hole points. The configuration consists of the frame and a set $E_{30}$ of 97 free points.

9.1. **Hole points.** A unit vector $y \in \mathbb{R}^6$ is a *minuscule direction* if $\langle t, y\rangle \in \{0, \pm\frac{\sqrt{6}}{4}\}$ for every tail $t$; these are the directions of the 54 weights in the two minuscule orbits of $E_6$. The 72 hole points are $x(u, y)$, as in Section 6, for 72 pairs $(u, y)$, with 36 distinct vectors $u \in \Lambda$ of squared norm 48, all of shape $(\pm 4, \pm 2^8, 0^{15})$, and 18 distinct minuscule directions $y$. The following facts were checked by computer:

(C5) for every hole point $x(u, y)$, every triangle not perpendicular to $y$ (a triangle is perpendicular to $y$ if all three tails are) carries a head set with $\langle u, h\rangle \le 16$ for all its heads $h$;

(C6) the hole points have pairwise cosines at most $\frac{1}{2}$.

**Theorem 9.1.** *This configuration is a kissing configuration of* $183888 + 38016 + 72 + 97 = 222073$ *points in* $\mathbb{R}^{30}$.

*Proof.* Pairs among the body and the lifts are covered by Proposition 2.1, and pairs with a free point by Proposition 6.5. It remains to check the pairs of a hole point $x(u, y)$ with the body, with a lift, and with another hole point.

*Hole, body.* The inner product is $\alpha\frac{\langle u,b\rangle}{\sqrt{48\cdot 32}}$. By (3), $\langle u, b\rangle \le 24$, so the value is at most $\sqrt{\frac{2}{3}} \cdot \frac{24}{16\sqrt{6}} = \frac{1}{2}$.

*Hole, lift.* The inner product is $\frac{2}{3}\frac{\langle u,h\rangle}{16\sqrt{6}} + \frac{1}{3}\langle t, y\rangle$, and $\langle t, y\rangle \le \frac{\sqrt{6}}{4}$ because $y$ is minuscule. On a triangle not perpendicular to $y$, (C5) gives $\langle u, h\rangle \le 16$, so the value is at most $\frac{2}{3\sqrt{6}} + \frac{\sqrt{6}}{12} = \frac{7\sqrt{6}}{36} \approx 0.4763$. On a perpendicular triangle, $\langle t, y\rangle = 0$ and $\langle u, h\rangle \le 24$ by (3), so the value is at most $\frac{1}{\sqrt{6}}$.

*Hole, hole.* This is (C6). □

## 10. Dimension 31

The frame is a Leech lift without poles: the 126 tails of type $E_7$ in 42 triangles and 42 head sets on the octads of Table 5. The partition of the tails into triangles, the head sets and

TABLE 7. The exact all-pairs check.

| dimension | points | pairs | result |
|---|---|---|---|
| 28 | 205172 | 21 047 672 206 | pass |
| 29 | 210495 | 22 153 967 265 | pass |
| 30 | 222073 | 24 658 097 628 | pass |
| 31 | 241079 | 29 059 421 581 | pass |

their assignment were chosen together by a constraint solver. The configuration consists of the body, the lifts and a set $E_{31}$ of 167 free points.

**Theorem 10.1.** *This configuration is a kissing configuration of* $174384+66528+167 = 241079$ *points in* $\mathbb{R}^{31}$.

*Proof.* Pairs among the body and the lifts are covered by Proposition 2.1. Pairs with a free point are covered by Proposition 6.5. □

Theorems 7.2, 8.1, 9.1 and 10.1 prove Theorem 1.1.

## 11. VERIFICATION

What is proved by hand and what by computer. By hand: Proposition 2.1, Lemmas 2.2, 4.1, 4.2, 4.4, 4.5, 6.1, 6.2 and 7.1, Theorem 4.3, Lemmas 6.3 and 6.4, the inner products of the tail directions in Sections 7 and 8, which are short computations, and the case analyses in the proofs of Theorems 7.2 to 10.1. By computer, in exact arithmetic: the facts on tails and poles in Section 3, the subcodes of Table 4 and the cosets $\gamma_i$, the disjointness of the head sets, the values $m_1(j)$ and $m_2(j)$ in Section 7, the facts (C5) and (C6), and the inequalities (F1) to (F4) of Proposition 6.5. These finite facts concern explicit sets printed in the paper or listed in the coordinate files. The all-pairs check below does not use these reductions. It is an independent confirmation, and it is the distributed program with which a reader can recheck every computed fact about pairs.

The paper determines the frames only up to choices that we give in the coordinate files: the coset assignments $\gamma_i$ and the translation labels of the 512-point parts, the partition of the tails into triangles in dimensions 28, 30 and 31, the assignment of head sets to tail sets in dimensions 29 to 31, and the 72 hole points. The free points are printed in Appendix A.

Each configuration of Theorem 1.1 is given as a list of vectors whose coordinates are integer combinations of $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$; the configuration consists of the normalized vectors. Both sides of (1) then lie in $\mathbb{Z}[\sqrt{2}, \sqrt{3}]$, and the sign of an element $a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}$ with integer $a, b, c, d$ can be decided exactly by squaring twice.

*All pairs.* One program, written independently of the programs that built the configurations, checks each configuration on all pairs, with no symmetry reduction and no floating point in any decision. It confirms the number of points and that every point is nonzero. For every pair of distinct points $x, y$ it decides (1), that is $\langle x, y\rangle \le \frac{1}{2}|x||y|$, by exact sign tests in $\mathbb{Z}[\sqrt{2}, \sqrt{3}]$, with 128-bit and 256-bit integers for the intermediate values; a pair whose values leave this range would count as a failure. It also confirms that no two points have the same direction. Table 7 gives the results. Every pair passes, and in each configuration the largest cosine is exactly $\frac{1}{2}$. This check uses neither the structure of the configurations nor the reductions above, so it confirms Theorems 7.2 to 10.1 independently.

*Structure.* Separate programs recovered the body, head sets and lifts and checked the 528-point head-set structure, subcode cosets, disjointness and matchings described above. The

layer and hole heads lie in $\Lambda$. A second exact program, written independently of the first, confirmed for dimensions 29 to 31 that every point outside $E_n$ is one of the frame points recovered and checked by the structure programs, and decided every pair containing a point of $E_n$ in $\mathbb{Z}[\sqrt{2}, \sqrt{3}]$; it found no violation and confirmed that each configuration spans $\mathbb{R}^n$. A third program, which we do not distribute, checked the reduced inequalities (F1) to (F4) directly from the coordinate files. The distributed all-pairs program decides every pair that contains a free point, so it also reproduces Proposition 6.5.

## 12. Conclusion

We have shown that $\tau_{28} \geq 205172$, $\tau_{29} \geq 210495$, $\tau_{30} \geq 222073$ and $\tau_{31} \geq 241079$, improving the best published lower bounds of 204522, 209497, 220442 and 238354.

In the Leech lift with the tails of [8], $\sum_i (|T_i| - 1)$ is already at the bound of Lemma 2.2. We do not know the largest head set. Each additional vector in every head set would add $\sum_i (|T_i| - 1)$ points, that is 16, 26, 48 and 84 in dimensions 28 to 31.

Layer and hole points replace some poles. The gain of layer points is limited by the head sets: a layer point needs head sets with small inner products against its vector $v$ on the triangles near its direction. Lemma 4.4 gives such head sets on the octads through the tetrad of $v$, at most 20 of them by Lemma 4.5. In dimensions 29 to 31 the free points replace all poles, and there are more of them (63, 97 and 167) than there are poles in [8] (40, 72 and 126). We cannot describe them in closed form, and we do not know how far they can be pushed.

## Appendix A. The free points

Each line gives one vector $z \in \mathbb{Z}^n$; the free point is $z/|z|$. The first 24 entries are the head, in the coordinates $0, \ldots, 23$ of Section 4, and the rest are the tail, in the tail coordinates of the coordinate files. The rows are in the order of the coordinate files.

**Dimension 29 (63 points).** $z_1$: 0,0,-2469,-706,2999,-1438,-867,-1246,1535,3471,0,-529,-3758,-1880,4605,-1059,24,-250,909,0,-149,1514,791,-1186,-15675,-15630,29496,-15140,-15154
$z_2$: -290,-13823,-6584,96,98,651,-98,-96,-3769,4559,-13823,-3891,-680,4492,64,3310,-344,-98,-88,-1,-51,-91,680,1091,16830,-16828,-16005,16830,16830
$z_3$: 1175,1314,-5329,2096,1729,-1806,-745,473,-704,4921,6450,3092,-2912,-3086,664,-428,-4006,-5489,1433,-257,-7094,4340,6225,2412,-18240,18222,-15604,-18855,-18317
$z_4$: 1125,-1759,3488,4001,-6425,839,3081,-4195,-2368,808,4313,-2868,9683,-1592,-4036,3963,-1174,-2986,-881,1274,2956,-4640,-3703,1344,18630,-18207,15444,18108,18228
$z_5$: 173,20321,10198,-10167,-10190,10320,29,3,-45,116,61,-38,-69,-10157,90,-106,36,83,-192,10227,-135,11,10180,10154,11942,10410,11643,11829,-11268
$z_6$: -5,20520,-10255,10244,-10275,10232,26,-17,-14,-13,-26,-18,-8,10243,-10,-1,-3,-16,26,-10247,10,1,-10235,-10239,12180,7110,12075,12130,-12080
$z_7$: 0,-20533,10266,-10266,-10266,10266,0,0,0,0,0,0,0,10266,0,0,0,0,0,-10266,0,0,-10266,-10266,10000,10840,13468,10840,-10840
$z_8$: 0,-10266,0,0,0,0,1,0,10265,-1,0,0,10265,10267,1,10265,0,0,0,-20532,-10266,10267,0,-10265,9467,11269,11468,-12841,11003
$z_9$: 914,635,-4955,-441,-147,-235,476,-550,5636,217,11741,536,774,-6105,170,-4231,-807,-1211,2116,-12144,-380,-961,3154,791,17399,18016,-14883,-17632,17917
$z_{10}$: -8092,0,107,-31,60,-16,-8549,45,8565,137,-16412,8733,76,8489,228,8733,31,45,138,8015,-31,-8457,45,31,14539,14539,-14540,-14540,14539
$z_{11}$: 19993,8912,-10207,-10308,88,-10556,213,217,0,210,-8912,-10267,-141,10068,10193,88,-216,-261,407,0,83,-206,261,282,11891,-11891,11891,-11891,-11891
$z_{12}$: -19829,-8956,10074,10128,126,10534,288,-95,266,288,8956,-10331,203,-10394,10138,-298,298,-180,-41,0,-279,-577,117,374,11929,-11929,11929,-11929,-11929
$z_{13}$: 19484,9049,-10154,-10343,94,-10021,-76,183,-281,-208,-9049,-10626,361,10154,10301,-96,-566,-370,523,0,525,58,-319,543,-12008,12008,-12008,12008,12008
$z_{14}$: -19872,-8944,10203,10597,266,10253,-102,-123,226,50,8944,-10275,226,-10109,10162,-176,142,-431,50,0,-255,-562,-165,213,-11920,11920,-11920,11920,11920
$z_{15}$: 8925,0,-291,49,-337,94,10266,87,10116,-351,19947,10720,-427,-10123,-31,-10067,174,-215,-156,-8925,292,10307,11,0,-11900,-11900,11900,11900,-11900
$z_{16}$: -8922,0,213,-117,97,-31,-10072,-158,10072,117,-19949,10679,6,10248,350,10411,240,117,31,8922,-371,-10116,209,158,-11902,-11902,11903,11902,-11902
$z_{17}$: 13,-10225,5,-72,16,-53,1,26,10103,-36,-3,-86,10223,10205,-127,10127,7,24,26,-20465,-10134,10097,-36,-10160,-9719,-12152,-11910,11453,-11792
$z_{18}$: 10,20522,10264,-10235,-10237,10248,11,3,15,6,-18,9,-12,-10266,-7,-8,15,2,1,10258,9,23,10218,10203,-7886,-11958,-11926,-11782,12341

$z_{19}$: -29,20394,-10218,10159,-10225,10224,-33,-46,73,-17,-77,36,74,10257,2,48,-23,-20,-29,-10104,24,-82,-10256,-10190,-9435,-12019,-11712,-11961,11665
$z_{20}$: 33,-20055,9844,-9987,-10229,10070,-220,104,-79,-167,-7,10,62,10066,161,67,149,-139,-113,-9851,64,-316,-10030,-9736,-10205,-12054,-12083,-12148,12521
$z_{21}$: 5,-20511,-10263,10249,-10265,10269,11,-1,3,-16,3,-5,10,-10260,-3,-3,-9,3,-5,10263,-6,4,10260,10262,11202,9606,12993,11002,-11317
$z_{22}$: -87,-20259,-10212,10109,-10333,10341,-109,-55,-240,-132,-77,-24,75,-10105,59,141,4,65,-97,10191,-30,-62,10270,10175,-10980,-11552,-11343,-11559,11693
$z_{23}$: -2,10266,3,-2,4,0,0,-2,10264,-4,-7,-3,-10265,10268,2,-10261,0,-5,0,-20520,10263,-10268,-3,10265,10604,10593,10945,-13359,10573
$z_{24}$: 1010,402,-1732,5417,-5907,-1652,2804,2534,-2621,-5976,-5296,-4826,298,1419,-5799,1499,-843,-1702,-2261,5883,-5258,-993,-1681,-2912,19502,17903,14140,-18769,18575
$z_{25}$: -2,10152,130,-17,-34,-17,10,-36,10099,-58,-59,-223,-10000,10206,-148,-10161,141,236,112,-20403,10118,-10295,-128,10237,-11266,-10973,-12216,11693,-11105
$z_{26}$: 224,260,-3764,4295,1811,1538,-1520,-1266,1249,-211,63,4817,-5838,-2785,6775,471,-3449,-3962,-4271,4003,-788,2998,5360,-280,-18901,-18567,-14528,18924,-19293
$z_{27}$: 650,284,9914,216,117,143,163,9950,35,9884,-9890,10049,-10086,-9853,-16,79,-425,-9439,-85,-10412,-9939,-9985,-10082,49,11259,11720,-12052,12440,-11982
$z_{28}$: -2,10,-10251,23,-7,42,-32,-10269,-15,-10267,10238,10198,-10247,-10251,2,-8,22,10256,25,-10244,-10244,10241,-10239,-34,10657,11498,-12808,10509,-10902
$z_{29}$: 10008,1083,-524,640,-342,4408,275,-3893,-3714,-385,1494,-1233,1449,695,5635,1346,406,-268,-3730,9764,-680,-901,519,1176,18605,19179,-11505,19084,-19625
$z_{30}$: 3,-1,10265,3,0,-2,1,10266,2,10267,-10263,10263,-10263,-10264,0,0,0,-10262,-2,-10267,-10263,-10264,-10264,-3,-10857,-10861,13450,-9990,10859
$z_{31}$: 11,-12,-10245,23,28,-8,0,-10237,35,-10212,10245,10258,-10248,-10282,-8,-11,1,10252,-2,-10258,-10254,10238,-10217,-17,-11972,-11789,12187,-12061,7921
$z_{32}$: -2233,2727,4599,1196,-8828,-170,-675,368,-439,-7512,1585,869,4363,3276,1693,3757,-23,299,-2352,-1122,2556,-7082,-310,-1447,-18448,-18218,15519,-19150,18248
$z_{33}$: 2,-10,10256,-4,-1,6,-6,-10264,5,-10261,10263,-10259,-10265,10266,2,-1,5,-10259,0,10257,-10264,10261,10263,6,12719,-9704,-11451,-10858,-11409
$z_{34}$: 0,0,-10264,-1,0,0,1,10267,2,10264,-10265,-10265,-10266,10265,0,-1,-2,10268,-1,10264,-10266,-10264,10266,-2,10423,-11331,-12962,-11157,-10232
$z_{35}$: 0,-14310,-3568,-37,-53,-7355,-21,37,3707,-3664,0,-27,-3599,3520,145,-27,-14,-3577,-16,-14303,37,-87,-26,-82,16493,-16493,-16405,-16493,-16493
$z_{36}$: 0,0,10266,0,0,0,0,-10266,0,-10266,10266,-10266,-10266,10266,0,0,0,-10266,0,10266,-10266,10266,10266,0,-13768,10540,10540,10540,10540
$z_{37}$: -1,-1,-10265,1,1,1,2,10265,-1,10265,-10264,-10265,-10266,10265,2,0,2,10265,2,10264,-10268,-10266,10262,2,-13550,10568,10617,10550,10723
$z_{38}$: -390,13295,3745,551,0,-344,-895,0,3573,-8612,-12951,-1586,1381,341,86,0,555,-344,551,-419,895,509,555,-3982,-17661,16882,16195,16882,16882
$z_{39}$: 0,0,4257,170,-888,3292,-386,704,-1157,-1871,0,109,-1940,1887,-427,1806,-445,2755,-1864,-3,-3663,-845,-891,-171,14982,-14616,-29603,-16225,15772
$z_{40}$: -7,7,-7,-2611,195,2390,-87,-214,1439,491,-7,2129,-5261,-500,3502,391,457,-3106,-1169,7,4,685,516,3075,-15202,13854,29417,16049,-16479
$z_{41}$: 665,-770,-2107,4067,9060,-6637,1716,2612,-2166,2592,-377,-104,-1236,-674,-726,-1609,-498,-1493,3982,-733,2424,9396,1785,-410,-19078,-18371,-15142,-19181,17582
$z_{42}$: 2447,-1791,3820,-1659,-1207,-1253,-493,763,677,1761,-668,5491,-5000,-2603,-5699,-2003,3979,3413,595,267,-5701,4482,8897,12,-18228,18169,-16571,19134,17411
$z_{43}$: -74,74,2304,177,2966,512,280,2295,-114,-1171,-144,-910,538,552,1096,1364,-5158,945,658,74,328,-1646,-5549,-74,15613,16191,-29456,15338,14238
$z_{44}$: 0,160,-316,-2019,4885,36,2343,509,2939,1461,0,1829,-1134,-816,128,-1256,-1444,154,2096,0,-512,2704,575,1853,-14034,15137,29698,-16739,15528
$z_{45}$: 110,-16,-1266,-3566,1498,-887,-356,279,923,4445,16,-1344,-148,-525,3316,317,1499,-1847,-2452,-16,1065,1350,398,803,15601,15571,29780,-15456,-14845
$z_{46}$: -308,159,2613,-171,-2779,184,622,1316,764,-162,-179,502,-590,1827,253,252,3509,3982,-2035,160,364,23,-2832,-1330,15281,16190,-30000,-14721,-14918
$z_{47}$: 0,0,510,-1787,3691,4533,-341,45,869,-613,0,1798,-933,1976,3371,-380,1750,1133,-2422,0,404,1679,1781,1244,16515,15326,29527,14562,15196
$z_{48}$: 0,0,2508,-408,-1833,1143,1176,983,274,-2735,0,-1597,5002,2058,728,883,-249,-58,-357,0,1356,-2396,-3574,762,-16182,-15795,-29553,14892,14729
$z_{49}$: 0,0,-252,-432,1403,-809,-268,871,1762,837,0,-1727,3525,906,677,899,-1146,2233,-58,0,3603,-3882,-4308,596,-15211,15577,-29515,14828,-15983
$z_{50}$: 0,0,-2657,-5028,1930,-2574,-1295,-18,499,1936,0,-1834,-1799,497,2023,-1429,-71,2168,1796,0,796,2153,796,-224,15583,-15485,29574,-14955,15576
$z_{51}$: 15024,14476,-3825,0,0,3825,0,-7649,-3825,-3825,0,-3825,0,3825,0,-3825,0,0,0,0,0,0,0,3825,-16049,16049,16049,-16049,-16049
$z_{52}$: -112,112,-828,-472,-1104,2659,1046,228,-373,-2050,-112,-441,492,2450,842,2287,1715,833,-520,258,2328,-4040,-3464,1216,-15655,15251,-29917,-15181,15142
$z_{53}$: 0,0,-459,-860,442,-73,-596,-139,505,95,0,-1761,-5702,743,5043,-688,-169,-105,-1356,0,113,891,2159,-3038,15644,-15539,29477,15762,-14655
$z_{54}$: 5862,5981,4472,4177,-882,3449,-1175,508,-3301,-1269,6156,5221,1657,978,-5407,3654,-813,331,2142,-6433,-2593,-204,-2176,3247,18132,-18600,15812,-18048,-18265
$z_{55}$: -2,13442,-2187,-469,94,1494,-465,-400,5903,-3382,13373,3475,-879,-4081,-331,-6380,-117,-465,396,-913,-400,241,603,-2875,16934,-16841,-16257,16934,16837
$z_{56}$: -4,-2,3511,578,562,182,97,1762,-430,600,-2,-1168,250,2345,-4624,-179,704,1607,2163,-45,-1667,-954,-1624,-1939,14897,-13864,-29735,16314,-16474
$z_{57}$: -7,-7,-4,-2438,1831,2026,-741,-849,490,373,-176,-237,-4719,-1222,189,-939,519,-33,-2344,-7,-3259,2540,1453,2701,-14583,-16593,29688,14951,15288

$z_{58}$: 545,-208,1312,-291,-1709,-1774,701,1241,-5062,79,1312,-4882,9529,274,-8741,-1815,-2146,4766,2716,3494,153,-2145,-3571,-4356,-17682,-18266,16089,18465,-18366
$z_{59}$: -11407,-1550,-1058,0,-500,-6373,-992,3439,3481,-492,-5526,1002,1251,-1614,-6444,-2770,-1233,661,5758,-4182,28,-2921,-19,1266,18540,17979,14755,18674,-17832
$z_{60}$: 0,-14180,-4022,-32,-32,-25,38,5,-4060,8143,14172,-5,48,47,-5,10,-43,40,30,0,23,-32,-9,-4116,-16624,16624,16598,16624,16624
$z_{61}$: -12273,-12427,7314,2503,-1107,3316,685,575,309,-3945,-949,2735,1261,-3985,-504,3883,674,828,-172,0,360,-380,-18,-2217,-17160,17046,15288,-17772,-17975
$z_{62}$: 87,-498,3256,-2195,-1752,2921,-1205,125,-473,1275,87,-1379,-308,1890,-1281,466,987,3190,-1400,-87,1822,-3091,-2414,-1731,-15617,-14788,-29911,-15021,-15587
$z_{63}$: 3,14094,-2246,-5,21,-2266,18,1,-2152,-12,-2,-17,6514,-6520,31,-23,-14,-2136,-20,14100,-15,-14,24,2195,16702,-16684,-16664,-16683,-16697

**Dimension 30 (97 points).** $z_1$: -363,4,-508,-410,-575,-1166,1833,-619,-829,-849,-15,-564,-558,91,-2336,155,-189,-1071,-982,73,759,668,-1362,342,14026,8931,1220,-25283,11319,-880
$z_2$: 1433,924,-136,406,-156,-667,-196,862,-326,-437,-938,-317,-531,-1959,-432,-845,-2109,457,924,-389,188,-915,-833,500,14622,6097,-729,25943,-10853,2079
$z_3$: 1366,895,11,437,-311,238,-1446,256,-587,-40,74,1175,1012,1101,828,856,96,915,-441,-1031,-393,479,575,-481,-29423,-13326,906,-947,263,2464
$z_4$: -100,304,-680,304,-511,-57,-1991,1341,-989,211,2793,150,118,-2495,151,-1420,-2149,5047,1218,271,566,-58,-1075,-1441,-7047,13764,12727,-8468,-23083,414
$z_5$: 753,505,138,299,-1515,179,-1551,-782,438,565,908,3304,3776,-562,3447,578,36,1648,141,-2228,-2483,2625,4135,-618,-7623,13312,-4273,12587,22555,-7778
$z_6$: 1128,-80,-1209,-2634,-51,1924,-1372,-990,151,931,-1195,2637,2613,-762,4010,-335,1238,2278,1283,-1588,-947,304,2223,87,14103,-26338,-5324,-1529,640,8669
$z_7$: -221,-118,417,-54,147,0,1339,483,855,-170,350,492,-483,-2,328,-376,1336,-155,189,425,-748,-277,361,353,4180,-10059,14511,2139,-248,-26909
$z_8$: 153,327,626,855,875,-67,419,-122,187,485,-79,111,776,-511,577,231,-934,425,641,132,-868,-590,168,-319,354,2907,15582,2146,9689,26616
$z_9$: -34,188,359,0,1469,223,1274,176,176,-597,-1691,-768,-2841,752,-1297,223,-399,585,-26,975,-625,-379,-1845,724,-648,6446,-29122,-406,-12258,-371
$z_{10}$: 371,-875,-887,619,362,43,-277,-473,-466,873,102,-763,-205,425,-331,730,1641,-1329,872,369,-1601,-564,-630,423,-14776,-8230,108,25619,-9741,3701
$z_{11}$: 1069,-247,126,621,445,313,2237,222,-1005,229,-1365,-889,-259,137,-1345,-368,-174,-696,-7,1990,-45,518,-245,72,-14990,-5321,2231,-25709,11273,-1101
$z_{12}$: 1114,-64,-680,-181,1072,-123,451,1419,925,-1025,-668,-744,-247,-527,-1246,-218,-1163,1077,251,-885,-1276,-215,-2881,-75,28872,14374,-1085,212,80,-92
$z_{13}$: -842,-1261,-2430,-2223,1785,-404,-1605,62,-442,966,404,1321,3057,-1540,-512,-1075,2593,-1785,2914,2044,-1914,2126,2496,-1017,6587,-12827,-11603,10946,23003,2
$z_{14}$: -120,252,386,332,-952,-244,-2319,-316,-777,1257,244,1893,610,1570,2268,-822,-1624,-1191,1242,-1388,-2119,4486,1809,1930,7270,-13811,6350,-11037,-22940,8507
$z_{15}$: -1457,-1763,-913,899,-234,1292,1652,1455,-692,461,1653,787,525,2270,103,-714,1581,-132,-301,-570,895,2622,2818,40,-13825,26684,7294,2448,-249,-7770
$z_{16}$: -129,31,-6,-955,-973,-28,-161,42,238,162,47,8,243,-48,139,-193,42,-618,-168,140,964,-18,-11,128,-3146,9029,-16057,-2340,-1360,26547
$z_{17}$: 567,137,-97,0,-635,432,219,20,317,-92,350,446,1605,-312,1605,-78,-10,327,-81,-105,-1,-37,876,-761,-1054,-5515,-15564,-2000,-9262,-26326
$z_{18}$: 302,-318,-633,-145,188,286,-94,24,-816,422,-547,-850,-1787,172,-684,409,607,-840,-106,-573,543,-648,-1139,1310,3818,-5611,30000,3562,9693,-672
$z_{19}$: -2,-290,-568,134,85,-62,-386,-212,215,-322,326,-861,-680,-199,-296,466,44,-184,128,-83,22,-724,-961,441,8647,20240,15069,14875,10799,3351
$z_{20}$: 49,729,136,-402,-198,585,-191,-427,895,-1301,-47,798,-251,-94,212,205,-1133,771,-585,-864,789,-267,141,-155,-8309,-16532,-21368,9759,1377,12664
$z_{21}$: 377,-20,127,5,-155,159,-273,80,-306,98,33,217,16,234,-7,451,268,-289,-296,-215,-83,453,522,264,135,-3879,6722,-23437,-13558,-16427
$z_{22}$: 448,-70,-27,-446,-14,-520,454,840,-22,-1497,-422,43,-1612,-641,-266,782,-1185,770,-1227,-616,-493,-1800,-1972,177,21190,-12962,-8473,12420,663,-14221
$z_{23}$: 948,-533,-468,-490,-718,-520,652,-127,177,-240,-212,-669,-1034,858,-817,130,924,-555,-532,-188,931,-914,-864,819,-12971,-8089,21236,2819,-13427,13466
$z_{24}$: -253,863,-374,482,1160,-644,195,187,-217,-955,-689,-879,-1930,-650,-1833,209,-1269,227,115,393,-398,-245,-2299,485,-8996,21017,-12776,-12567,14105,1283
$z_{25}$: 2019,-226,226,-577,-4280,-19,-1764,313,-6,-579,2028,1253,4389,305,-21,556,750,-377,343,-2116,-535,-1114,4362,-635,12031,11890,-7741,-82,-22604,11394
$z_{26}$: 2639,-247,-346,202,985,1314,-5335,-305,-3357,2362,-1302,-1022,837,-1337,-499,357,-895,-202,945,-3413,1596,343,368,-1181,-18716,754,1793,12873,-129,-21599
$z_{27}$: -878,-302,-520,1144,-1308,3258,-1048,1602,2358,-795,1929,592,1070,-983,1942,-456,997,-1723,794,29,-943,1475,1707,-1179,7506,-13236,9311,-12319,21607,9009
$z_{28}$: -396,-243,314,531,513,-324,806,88,99,0,-230,-1844,-617,314,-1471,299,-955,-104,-442,543,182,-1517,-2253,-131,-6655,-22190,-13806,-14061,-10812,-2493
$z_{29}$: 142,-226,-71,498,-561,244,351,351,-207,-375,-74,-81,-647,10,-338,-385,455,180,256,-71,351,-429,-125,-910,10324,16079,20808,-8599,-2134,-13598
$z_{30}$: -116,-82,861,438,168,639,460,-97,437,-60,270,-322,-438,484,-120,536,-53,-248,-531,72,-40,-377,-1053,-21,-1092,4958,-6579,23565,12164,16974
$z_{31}$: -1420,523,-750,-902,-468,-186,-1197,-70,1129,192,546,984,464,-694,492,69,127,-149,0,-2003,308,-690,9,317,-21697,12308,6320,-11556,-714,15966
$z_{32}$: -426,-231,-585,74,280,513,-140,81,559,-27,575,741,316,972,25,-224,134,358,-158,-2029,-1553,303,1321,659,12111,7150,-21936,-2218,13453,-13700

$z_{33}$: -362,-160,-1122,83,-229,12,-167,-235,370,-17,-1142,-825,213,-694,-464,196,-467,-937,-434,747,359,-58,-1685,-1136,9441,-20143,14797,14069,-11814,-1687
$z_{34}$: -868,-322,-338,37,-1487,-2863,-874,-1119,-1933,19,2953,887,2877,512,1775,-937,-1858,2942,472,967,1180,-799,2076,-1341,-11313,-12001,10049,71,22058,-12024
$z_{35}$: -2038,-1143,1233,1606,-910,-3685,-3635,-2500,-1962,-123,474,1745,817,1335,389,-881,-1213,-972,-664,-126,1561,815,2037,-699,19783,-1342,-224,-11934,-1325,21513
$z_{36}$: -540,-395,136,-94,-1300,377,-716,14,78,907,1070,1334,1941,-1648,4156,-875,785,-2339,1775,-2682,422,1521,3875,-346,-7993,12959,-7334,12991,-21433,-9265
$z_{37}$: 151,-1352,-860,-882,-2733,-1026,1126,-467,616,317,1047,-415,2551,-894,1197,206,-707,-1264,-479,-1211,71,2962,3282,-249,5381,25632,-14036,11146,750,3112
$z_{38}$: 3162,813,2537,1876,-2087,-913,1137,573,-728,1032,210,559,-624,218,-85,-1760,-1114,2087,780,-1164,2077,1090,3954,151,-7012,-25408,-4733,11175,668,-12676
$z_{39}$: 1191,1068,-1056,877,-452,-283,-3948,-598,-947,1789,2151,1158,2834,-722,2447,873,783,1205,-634,-1775,154,-581,-313,-378,1144,376,21346,-21295,-2380,9812
$z_{40}$: -741,192,593,589,-681,-631,535,-134,-706,-54,239,-977,-775,859,-898,840,31,-211,-1919,293,1154,-438,-60,364,22543,-7572,8384,13490,11025,10637
$z_{41}$: 1902,576,318,395,899,729,-929,-28,-1541,94,-476,-54,-206,454,-1307,417,-71,1157,455,-729,-916,-189,-1130,-439,-16316,-2121,-8485,-254,-23387,12628
$z_{42}$: 687,682,389,-170,31,-786,-726,-553,1133,610,-237,-113,2102,64,1291,1334,-602,-538,-272,227,-1171,285,1391,970,-6922,10084,1487,-12087,11628,-24787
$z_{43}$: -845,-182,-196,153,-3,4,26,-556,-318,-19,-685,-345,-363,531,2,359,-7,-773,-183,661,-7,743,-506,37,17035,3137,3069,2332,-22265,-15809
$z_{44}$: 575,193,-390,-162,1217,-173,325,-568,-46,-577,37,-619,-1634,-649,-642,-1011,-319,-245,28,597,956,-600,-1748,-605,-20396,5830,15640,14297,11176,5284
$z_{45}$: 113,-526,564,375,714,-494,1792,-15,-386,-896,-688,-1053,-1133,406,-1158,-304,108,76,-808,22,162,-467,-2408,206,6987,-8419,-20309,-14320,13801,10962
$z_{46}$: 94,296,177,-985,-363,620,583,118,1053,-29,-747,3661,-1700,-384,728,165,925,-1563,683,1080,-135,896,3722,45,-5521,-24884,15478,-11050,-933,-3506
$z_{47}$: 196,45,1182,2684,2710,-860,-509,-58,806,365,-98,1393,365,1202,1466,1,-493,2743,1248,605,-2699,1031,4687,22,6391,25517,6593,-10262,-1906,12528
$z_{48}$: 364,-1242,298,-529,-4089,-267,-8,517,-100,2322,3141,-160,1182,596,1747,830,-1850,-1532,155,-365,2140,2976,486,-712,-2118,142,-20024,22161,2117,-10183
$z_{49}$: 679,797,38,-855,620,-1089,1169,-263,80,-437,-1453,-1087,-2548,48,-1768,292,-1160,828,-538,738,-136,-693,-1249,297,-22787,8119,-7332,-11342,-11175,-12219
$z_{50}$: 532,-190,-300,86,547,-426,666,185,354,-620,829,-66,1159,901,-1045,787,-1077,-214,130,139,-1473,831,-863,76,16580,2838,8631,288,22478,-13808
$z_{51}$: 144,-637,183,749,1447,-432,-190,528,-428,97,-454,-770,-1648,801,-315,362,1926,-23,707,-261,-1558,280,-672,51,6062,-10023,-282,12466,-12231,24638
$z_{52}$: -107,-410,-22,-288,-544,-56,169,-283,-494,782,789,227,642,509,740,33,51,404,659,0,-93,186,1310,-345,-15681,-2060,-5268,-2061,23414,15117
$z_{53}$: 548,405,119,-777,-678,910,-1051,910,712,977,344,1016,1282,-748,657,920,594,16,715,-149,-322,71,1630,-156,20544,-5832,-15666,-15137,-10344,-3615
$z_{54}$: -670,453,471,-187,152,-329,1093,41,397,87,-135,692,52,49,248,-478,52,2003,-439,581,-186,1284,0,-607,-4167,7194,21324,17633,-11806,-8991
$z_{55}$: 155,-126,392,-134,546,94,1166,781,341,-295,-505,-593,-1869,315,-851,-1205,915,363,-23,1532,967,-1170,-465,227,7620,16151,-713,13718,2029,-23197
$z_{56}$: -1637,-234,-371,-103,1480,237,-721,-383,583,-1411,331,-391,525,553,-1056,795,221,0,-1003,842,267,-631,-2338,-477,-6826,-22293,8680,11967,11464,12352
$z_{57}$: -332,-55,-106,140,1247,389,2224,56,1023,-1017,-2085,-1338,-659,-130,-1428,536,109,-442,-344,968,378,-1065,-1204,-207,-512,6341,-6423,-25535,-13684,11042
$z_{58}$: -1613,-27,-28,-1913,1847,663,-640,-1267,-1212,859,2018,1356,1037,2145,1771,-139,2967,-414,-737,-548,-407,2228,4247,1303,19191,-1349,-20126,9755,669,11412
$z_{59}$: -4185,-2440,-1064,-135,43,801,1582,-374,1361,-589,-322,-858,1403,-1042,1260,-1096,595,-3445,525,1952,-17,-69,586,71,-14013,-12707,5752,1983,-21526,-12373
$z_{60}$: -310,-1632,1058,-1741,-3357,568,-797,-715,1516,548,1760,1309,355,1545,2900,81,-1442,559,-2341,-2985,744,186,2393,-1279,-6005,13301,15748,-9996,20944,1580
$z_{61}$: 29,174,-180,-1419,1480,72,-203,-900,-111,87,75,8,-1468,296,61,-105,0,-76,-740,-104,-12,-31,-973,-116,16073,6173,20603,3443,-13577,11778
$z_{62}$: 523,-107,129,-61,-591,1288,-934,239,-725,543,1205,-115,854,-245,1014,-384,197,2274,24,-241,-357,709,1360,0,-23545,11984,-14339,10719,758,5518
$z_{63}$: -692,294,91,231,-832,-203,-459,147,60,261,-204,-124,-985,549,1107,-345,297,832,447,-646,478,-617,-560,-12,7965,-16007,-5495,-16719,11982,-16887
$z_{64}$: -570,1122,-347,-872,-1603,706,-1156,13,-54,190,553,595,-338,535,1596,-112,1603,715,162,-84,961,507,1569,360,-8340,-15556,300,-14162,-545,23127
$z_{65}$: 448,210,321,-41,-863,-412,-753,640,-234,-44,951,62,1566,-920,619,-847,1402,191,-106,-69,347,-360,596,-198,7036,21554,-9748,-13953,-9400,-12743
$z_{66}$: -390,896,-148,386,244,263,942,768,563,-212,-331,-800,-2406,-90,340,40,-1153,-567,-1226,-1397,-555,-1292,-1635,537,1505,-6388,8691,25721,12428,-10504
$z_{67}$: -3600,-445,1929,334,2522,493,-1240,1774,877,-2426,1013,954,2177,-1292,1338,259,770,-1679,123,-956,663,893,-413,70,-18853,1720,21604,-8773,-1380,-10481
$z_{68}$: -241,1642,121,-2052,-2045,1837,-1407,-507,-1507,2402,-2065,2252,1488,598,2130,-2015,1242,-2224,-208,-41,2063,-1127,4003,-567,13052,12294,-6088,-586,22503,11214
$z_{69}$: -670,-669,1430,334,-2844,313,758,1106,801,819,4269,2021,1648,-1010,514,-158,79,1519,-1974,4,1498,1039,4375,573,5447,-13857,-13279,10822,-21876,-1187
$z_{70}$: -269,43,272,-321,858,-279,628,283,175,-527,-808,-364,-1028,-177,-1262,-21,157,-857,36,713,-590,-548,-1173,-155,-17356,-5334,-18751,-1330,14646,-12592
$z_{71}$: 51,-494,333,253,-595,12,822,569,348,-341,918,618,1987,-827,-847,0,-100,188,504,959,-117,1446,674,334,24054,-11051,14437,-10857,-1281,-4857

$z_{72}$: 205,-2,324,385,978,-820,381,-89,-104,-90,-773,-258,-1179,498,-1049,511,628,-1663,138,776,576,-332,-219,265,-7592,17849,6121,14280,-12304,16901
$z_{73}$: 0,0,0,11541,11546,0,0,0,0,0,0,0,0,0,0,0,0,11495,0,0,11546,0,0,0,509,-3816,13282,-882,-15253,-10399
$z_{74}$: 0,0,0,11544,11544,0,0,0,0,0,0,0,0,0,0,0,0,-11497,0,0,-11543,0,0,0,620,1751,14701,-1074,-16704,-5750
$z_{75}$: 0,0,0,11544,-11544,0,0,0,0,0,0,0,0,0,0,0,0,11520,0,0,-11520,0,0,0,523,-3851,13285,-905,-15241,-10398
$z_{76}$: 0,0,0,11532,-11532,0,0,0,0,0,0,0,0,0,0,0,0,-11532,0,0,11532,0,0,0,535,965,15101,-927,-15933,-6986
$z_{77}$: 0,0,0,-11532,11532,0,0,0,0,0,0,0,0,0,0,0,0,11532,0,0,-11532,0,0,0,715,-3838,13186,-1238,-15164,-10593
$z_{78}$: 0,0,0,-11520,11544,0,0,0,0,0,0,0,0,0,0,0,0,-11520,0,0,11544,0,0,0,584,1441,14603,-1011,-16703,-6093
$z_{79}$: 0,0,0,-11544,-11548,0,0,0,0,0,0,0,0,0,0,0,0,11493,0,0,11544,0,0,0,683,-3865,13151,-1184,-15173,-10621
$z_{80}$: 0,0,0,-11531,-11534,0,0,0,0,0,0,0,0,0,0,0,0,-11530,0,0,-11534,0,0,0,607,1494,14615,-1051,-16714,-6013
$z_{81}$: 0,0,0,11524,11524,0,0,0,0,0,0,0,0,0,0,0,0,11529,0,0,11551,0,0,0,-866,12963,-2453,1123,10940,15397
$z_{82}$: 0,0,0,11532,11532,0,0,0,0,0,0,0,0,0,0,0,0,-11532,0,0,-11532,0,0,0,-1208,13043,-2880,682,10840,15329
$z_{83}$: 0,0,0,11521,-11554,0,0,0,0,0,0,0,0,0,0,0,0,11526,0,0,-11527,0,0,0,-844,12953,-2451,1131,10952,15397
$z_{84}$: 0,0,0,11555,-11539,0,0,0,0,0,0,0,0,0,0,0,0,-11517,0,0,11517,0,0,0,-1474,15964,2274,3,5736,15419
$z_{85}$: 0,0,0,-11557,11524,0,0,0,0,0,0,0,0,0,0,0,0,11524,0,0,-11524,0,0,0,-868,12949,-2475,1121,10954,15395
$z_{86}$: 0,0,0,-11521,11541,0,0,0,0,0,0,0,0,0,0,0,0,-11527,0,0,11539,0,0,0,-1496,15962,2239,0,5766,15412
$z_{87}$: 0,0,0,-11520,-11524,0,0,0,0,0,0,0,0,0,0,0,0,11546,0,0,11538,0,0,0,-795,15767,2559,265,6093,15486
$z_{88}$: 0,0,0,-11540,-11533,0,0,0,0,0,0,0,0,0,0,0,0,-11512,0,0,-11543,0,0,0,-1481,15963,2252,0,5761,15412
$z_{89}$: 0,0,0,11519,11539,0,0,0,0,0,0,0,0,0,0,0,0,11539,0,0,11532,0,0,0,1276,-13612,-14691,0,9842,-5763
$z_{90}$: 0,0,0,11554,11516,0,0,0,0,0,0,0,0,0,0,0,0,-11516,0,0,-11543,0,0,0,795,-15672,-12749,110,5256,-9817
$z_{91}$: 0,0,0,11532,-11532,0,0,0,0,0,0,0,0,0,0,0,0,11532,0,0,-11532,0,0,0,1380,-12865,-15398,0,9744,-5776
$z_{92}$: 0,0,0,11518,-11539,0,0,0,0,0,0,0,0,0,0,0,0,-11539,0,0,11532,0,0,0,1274,-13611,-14692,0,9842,-5762
$z_{93}$: 0,0,0,-11519,11539,0,0,0,0,0,0,0,0,0,0,0,0,11539,0,0,-11532,0,0,0,200,-15330,-13511,0,4660,-9670
$z_{94}$: 0,0,0,-11513,11538,0,0,0,0,0,0,0,0,0,0,0,0,-11538,0,0,11538,0,0,0,568,-15471,-13181,6,5014,-9708
$z_{95}$: 0,0,0,-11540,-11524,0,0,0,0,0,0,0,0,0,0,0,0,11524,0,0,11540,0,0,0,1278,-12436,-15863,0,9466,-5940
$z_{96}$: 0,0,0,-11518,-11539,0,0,0,0,0,0,0,0,0,0,0,0,-11539,0,0,-11532,0,0,0,198,-15329,-13512,0,4659,-9671
$z_{97}$: 286,-359,-700,-104,-4454,239,-4836,-46,-130,3780,4913,3707,3548,318,7293,-527,-326,358,534,-3924,47,3570,6844,118,-10131,629,-19856,-18154,73,-217

**Dimension 31 (167 points).** $z_1$: -111,35,514,-310,-139,-211,-137,-155,-64,-130,-22,354,671,79,56,-194,786,-547,443,-57,259,-57,263,234,-14123,-15038,62,-11315,10949,-10543,-11014
$z_2$: -86,-425,633,-532,224,119,-73,-289,-262,-408,-917,-343,152,495,-84,-329,-444,160,89,138,262,-821,497,394,15102,14909,-54,11007,10959,10601,-10019
$z_3$: -328,659,-1154,224,261,166,-500,4165,342,2468,-866,-1311,-5122,-1846,1325,116,1811,1322,-291,168,1459,804,-1029,-806,-837,163,-42,247,-20271,-3858,20310
$z_4$: -533,834,-816,-668,452,864,163,1182,-305,311,38,561,-970,-440,1410,369,1405,-21,-366,-824,-725,-1520,603,628,897,-15089,-14950,-326,485,237,-21009
$z_5$: 18,-620,1000,23,-21,-204,-558,-342,240,346,271,741,721,274,-629,-611,119,-638,605,-244,465,-81,-464,-480,190,15047,14885,49,32,21327,-18
$z_6$: -780,-1567,1539,184,-407,-141,-1662,-2535,-938,-924,-1355,695,2107,1392,-1428,-342,-2927,-419,-503,867,33,-2115,1066,256,-988,-175,37,111,202,-21185,20388
$z_7$: -198,337,-87,510,-145,-62,-70,-131,-59,141,344,325,416,24,155,201,252,-363,374,-390,-14,649,-788,-284,-14826,14878,160,10177,11007,10676,-11221
$z_8$: -129,-635,-202,23,686,-670,771,730,-641,1014,-477,-1069,-762,26,241,212,939,-1540,183,535,-231,1273,-782,93,57,-15007,15028,-267,-134,-21104,-64
$z_9$: -124,0,-145,-39,276,-89,-125,777,-160,662,30,29,-951,-166,419,-79,508,-33,-181,116,40,43,272,-341,15063,-11,-15282,-9878,-10739,10435,11106
$z_{10}$: -89,-130,380,-198,-578,-213,718,-200,134,-182,530,-215,-248,252,-589,-534,586,-516,-88,290,90,191,274,-28,15145,-111,-15080,11050,-10250,10595,-10443
$z_{11}$: -492,-97,997,76,-103,332,-284,195,-853,1493,545,825,179,-1037,-724,312,1146,-347,346,-57,-804,-18,-989,96,165,15001,15072,-35,21110,16,410
$z_{12}$: -52,673,11,-152,177,153,160,-487,-330,-1118,123,560,367,239,-218,522,-699,492,77,-455,-149,-625,804,-50,-15013,-15085,-10,-11097,-10916,-10824,9538
$z_{13}$: -79,64,278,444,-59,-103,0,141,-301,-240,303,71,481,369,171,-680,379,43,-48,89,239,-23,84,-145,-14309,41,-15112,-11129,-10230,-10583,-11537
$z_{14}$: -143,-91,-45,-43,-8,77,-148,156,-110,64,18,19,-109,69,-70,-102,-37,-181,134,37,-31,-52,-183,-42,-476,14928,15166,21327,89,-83,-458
$z_{15}$: -372,-265,138,-71,-35,0,-366,-654,-91,-206,-84,-168,792,-65,-264,-219,14,-242,185,106,-370,-100,161,445,15026,-15014,-78,-10123,10335,10506,11636
$z_{16}$: 2,-306,-1163,240,290,642,387,585,2,120,-26,-44,-1109,-607,120,-243,-288,431,-412,-235,104,226,-525,-170,15771,-14997,-42,-10430,10790,-10486,-9724
$z_{17}$: 363,-184,448,-178,364,-555,-1146,-305,91,-471,-414,-896,900,947,542,-518,318,-287,-980,447,-492,341,-1714,1195,74,29941,21,159,186,-92,-242
$z_{18}$: -117,672,-533,319,0,457,718,909,17,271,247,4,-1373,150,465,482,167,595,-260,-531,306,201,712,277,-15539,-15283,159,10325,-10962,10693,9266
$z_{19}$: -29,-243,112,-315,86,-495,-189,30,94,741,-322,-89,-507,-79,-422,-223,313,-254,-338,245,-239,69,-24,-190,-14835,-15138,104,9889,-10342,-10589,-11841
$z_{20}$: 12,-103,90,-16,65,-35,-139,171,-42,156,-86,-168,36,56,91,-64,69,-209,116,4,87,-29,-120,-53,773,15243,14935,-21262,-49,131,163
$z_{21}$: 62,-197,107,-49,202,-477,-277,-465,-257,-641,-248,-488,540,-163,-50,-100,-192,-28,159,96,-254,-37,237,128,14271,-96,-15077,11323,10504,10437,11314
$z_{22}$: -378,76,-387,-474,-246,458,461,316,-103,363,816,-329,-437,752,689,773,12,-220,-360,-497,-543,228,-213,-442,15717,15023,-73,-10228,10908,-10500,-9847
$z_{23}$: 0,0,0,0,0,-6,0,6015,0,3023,-3004,3,-6009,0,0,-19,0,0,0,3,0,0,0,0,21,6,-12,-21,-20060,20068,-3576
$z_{24}$: -237,509,-836,-225,-339,382,325,-19,-394,-634,-289,-388,-702,-330,538,735,-162,630,-462,-16,-9,-271,692,-13,-15603,-15314,268,10190,10471,-10765,9804
$z_{25}$: 721,-407,547,-54,111,-212,261,-489,-447,37,-119,-199,143,1,107,-105,272,-495,-102,42,-296,-206,250,-148,-14171,-34,-15009,-11040,11060,10747,-10972

$z_{26}$: -294,-720,91,28,238,-771,-368,15,217,-393,-286,348,449,759,-546,-709,-499,-165,184,358,598,4,150,-281,116,15134,
14900,20,-21272,-60,-741
$z_{27}$: 104,-280,748,92,-46,-492,-131,68,91,521,-26,-36,-44,336,-165,-209,-218,-187,231,-23,-104,-651,-129,0,14325,-15122,
47,11027,10400,-10872,11161
$z_{28}$: -516,-473,89,-417,-1348,81,-551,1508,62,1072,322,357,-1614,-165,201,525,361,-25,61,-490,1164,-660,1454,-1251,818,
14828,-14993,-60,602,234,-21152
$z_{29}$: 1346,-1059,817,434,-219,-714,-313,-177,324,-284,-444,328,345,524,67,-510,-474,-148,-668,-34,-317,1322,-357,-162,84,
-30000,83,-207,-29,-53,-40
$z_{30}$: 146,308,-411,-5,-236,148,168,-278,177,-362,-88,-32,-463,-325,326,345,-130,506,20,-47,-103,-86,429,-154,-736,15138,
15155,263,-606,86,21140
$z_{31}$: -1182,381,-755,55,-273,-22,-449,-267,752,-287,513,106,621,221,-183,168,-133,373,44,-492,611,-411,-846,240,15719,-7,
-15107,-10160,10909,10645,-9588
$z_{32}$: 84,-176,196,375,152,-526,468,27,-402,-211,145,-165,792,601,-93,-290,-899,-356,-188,154,264,484,-220,-302,121,
-14950,14953,-177,-21386,-20,-617
$z_{33}$: -190,644,-388,-270,-493,558,277,-60,-227,-650,60,56,-941,35,342,477,-82,899,-231,-191,260,-418,671,164,-15676,
15093,153,10402,10574,-10733,9745
$z_{34}$: -187,-306,-142,112,-186,227,-318,-221,267,928,-191,20,-180,102,-29,-132,-566,-801,-219,-6,164,53,-762,377,-14939,
-183,-14876,10080,11031,-10609,-11174
$z_{35}$: -281,-223,341,194,90,125,-631,-397,85,285,514,910,508,-335,-940,-494,420,-187,502,-286,-243,-241,-1030,126,200,
-15050,14972,-150,-128,21268,50
$z_{36}$: 151,-152,134,-260,-178,-157,82,-641,-211,-476,346,394,885,482,-526,-188,-701,-341,123,-42,-180,-75,-63,471,14939,
15228,-158,-9974,-10817,-10811,10797
$z_{37}$: 337,1020,151,-381,-7,-171,-302,362,258,-742,-191,253,206,-556,187,65,644,26,178,599,132,-569,545,324,15194,-108,
15050,10705,10858,-10558,-10141
$z_{38}$: -147,-343,-177,547,-639,-528,-200,-20,-324,-789,-738,51,137,-159,-379,-49,-896,48,-400,460,-187,658,-319,72,66,
-15072,-15001,64,92,21255,65
$z_{39}$: -132,306,-508,163,-551,247,547,-861,67,-731,367,491,387,487,115,691,-662,400,-288,44,-274,-356,736,334,-15084,
15299,-26,-10960,-10809,-10747,9444
$z_{40}$: -607,-324,212,-201,148,-32,-755,-281,291,116,-142,844,393,-164,-453,-314,789,-123,175,-48,-400,-334,-794,676,9,
14994,-15145,220,-21191,-72,-711
$z_{41}$: 106,-86,-3,-35,-21,-137,-162,-219,73,60,-16,115,277,0,-169,-143,171,-212,270,-90,-99,101,-273,-19,-467,-15215,
15270,21043,-37,-72,-412
$z_{42}$: 122,-55,112,83,-161,217,113,240,12,237,-162,8,-187,107,-37,-4,84,-64,-78,300,154,158,-46,-24,691,72,-57,-21185,
21402,63,759
$z_{43}$: 527,383,486,194,475,-308,250,547,-314,297,171,-213,-32,-266,998,-69,403,328,-208,80,464,-183,647,-229,15011,-156,
-15008,10894,10987,-10618,-10094
$z_{44}$: -833,1719,-2455,426,-125,1029,2501,-866,-870,-1673,-444,1120,-3756,-1087,311,2513,-1022,3130,-927,-456,-1711,-1630,
1717,-1523,-28782,-24,88,-939,221,-4,-3408
$z_{45}$: 22,-197,179,-1,-25,-13,-169,-115,5,119,-54,68,140,52,-189,37,-61,-159,-26,77,16,102,-43,20,613,-14937,-15218,
-21284,57,-4,250
$z_{46}$: 150,-686,1221,211,-325,-490,-679,250,565,773,198,458,175,10,-478,-931,970,-1293,117,15,140,774,-747,-17,214,71,
29987,-195,-188,15,-181
$z_{47}$: -51,-25,-56,-14,-29,-125,90,164,-137,-14,154,11,181,-14,-130,-125,187,-64,112,-13,-148,81,-248,-69,-643,14944,
-15035,21402,162,-111,-437
$z_{48}$: 245,-108,-158,97,-45,-240,-481,564,236,173,36,-53,642,21,161,-257,-487,-272,143,46,185,194,23,53,-14139,73,14986,
-11353,-10394,-10541,-11581
$z_{49}$: 61,-5,28,-40,101,-100,-121,-69,75,-88,-15,76,139,-89,-51,-117,117,-40,172,75,-89,63,-172,-133,-613,-15311,-14933,
21219,85,-100,-370
$z_{50}$: -39,-9,63,198,-122,-757,99,-841,95,-177,86,-63,800,-197,12,-108,-174,269,-12,52,-89,-187,-3,284,14984,15192,-149,
-9903,10463,10465,11562
$z_{51}$: 169,13,-85,246,432,-3,-224,-500,-359,-219,-145,-60,564,79,37,92,-155,226,26,-57,-142,-128,-381,-24,-14847,-119,
-14989,10209,-10228,10640,-11826
$z_{52}$: 92,-152,190,-45,-11,-170,-136,160,58,209,-60,-166,-9,120,-85,-4,53,-134,63,162,15,-60,-55,-23,769,-14927,14969,
-21460,-141,76,251
$z_{53}$: 484,-565,-74,43,410,156,-201,-288,293,-22,-174,-243,356,-3,-192,-409,-33,-700,-134,98,161,-47,-92,-1,14256,14999,
29,11224,10511,-10929,11061
$z_{54}$: 264,470,-359,-282,73,-84,245,1027,27,126,-258,222,-1396,79,247,212,677,168,32,-378,504,-677,513,-473,15952,67,
14813,-10347,10713,10832,-9470
$z_{55}$: 706,55,1032,-425,404,-315,436,-838,955,1129,1685,1003,-256,129,-467,-198,1253,-64,937,-500,279,349,-1000,-373,-122,
-14984,-14976,-93,92,-21124,-111
$z_{56}$: 361,786,-469,0,214,287,59,875,298,386,370,85,-1466,-409,587,471,740,681,-166,-85,169,89,314,168,-15466,15418,84,
10383,-10886,10648,9243
$z_{57}$: 525,0,-361,875,341,21,-194,-789,-28,68,-86,-119,71,-320,121,-424,421,163,-157,33,349,7,-72,144,15097,14978,-67,
10971,-10244,-10640,-10720
$z_{58}$: -162,-986,277,-479,-201,578,-1260,662,386,1257,-543,1578,-280,-829,-348,-368,1558,-699,498,-303,-129,36,-768,302,
209,-15057,15074,-255,20997,-35,515
$z_{59}$: -287,209,-261,117,-116,351,208,1234,199,485,349,-172,-904,-357,394,579,64,115,-137,-174,20,-53,903,-490,-15090,114,
-15525,-10687,-10676,10636,9664
$z_{60}$: -360,-94,268,-669,-232,-96,97,384,94,267,-178,-32,104,221,-9,57,-212,-178,546,-139,58,-107,-216,83,-14894,14953,63,
10127,-10241,-10602,-11899
$z_{61}$: -1131,-904,985,580,382,-229,-966,-2391,751,-490,-511,1685,1557,-1300,-1940,-713,131,-1228,2085,637,-342,124,-1720,
1008,28,-61,32,29,21127,20713,220
$z_{62}$: 79,-53,128,-133,-216,-164,159,-1007,290,-324,-109,174,1003,105,-265,-27,-276,-452,-142,62,-317,429,-25,180,14983,
-14933,-128,-10164,-10918,-10817,10874
$z_{63}$: 183,-126,-55,-392,141,96,429,-792,-203,428,-290,85,892,-345,508,-144,-586,176,-508,-92,645,387,220,367,15792,
-14892,-181,-10385,-10410,10714,-10112
$z_{64}$: 756,-3,-566,874,490,-721,1519,-432,-866,183,35,-1107,-169,-328,228,107,-1384,-681,-119,879,185,1299,-774,-668,-48,
14880,-14875,184,21270,-10,445

$z_{65}$: 0,394,-157,-188,8,405,45,-1408,19,-767,-193,275,177,-155,-44,858,-132,1359,-7,-426,-81,-1061,818,257,-15542,-39,15350,10197,-10875,-10855,9130
$z_{66}$: 129,1631,-1213,-481,-432,34,-3,-11,77,-890,297,159,-1845,-95,1061,707,148,1667,-340,-278,359,-510,1119,-46,920,-15091,14949,-492,330,250,-20994
$z_{67}$: 23,128,351,244,253,-185,-204,1074,901,-1473,694,906,406,644,-956,-478,-1143,1362,741,-717,206,71,-1540,-542,-130,14958,-14907,115,170,-21163,-207
$z_{68}$: 1298,1279,-230,-280,-408,1273,651,4056,-207,2828,-279,1146,-4632,388,832,1500,974,-557,-848,-968,1584,591,401,-1097,-565,66,9,247,2151,20644,20346
$z_{69}$: -55,-35,31,113,90,215,15,-407,455,-476,306,6,736,-5,-253,101,119,-292,5,249,-137,-172,-318,-191,-14217,-15004,-89,-11302,-10345,10720,-11389
$z_{70}$: 1977,516,434,-364,401,-334,737,738,-745,786,221,-370,-723,-373,83,743,515,943,570,453,-36,1861,-2385,-261,-32,-110,123,-94,21103,-21058,458
$z_{71}$: -120,196,-259,44,-138,268,-115,771,-44,388,-44,138,-914,-163,66,268,219,163,-286,196,116,98,259,-263,14577,15071,-166,11373,-10683,10360,10754
$z_{72}$: -421,283,103,-196,-473,313,366,148,149,-159,199,-96,-1095,-381,360,631,997,59,-68,153,216,61,152,-157,15818,103,14823,-10418,-10506,-10467,-10300
$z_{73}$: -322,0,161,-232,-240,24,-393,614,-24,-140,112,-94,136,195,-236,-411,-258,6,-53,858,414,-22,-437,459,-14750,-15170,198,9974,10908,10682,-11208
$z_{74}$: 49,-189,30,3,-170,63,-53,-21,-4,102,123,104,75,-40,-10,-47,-174,-125,15,-82,235,-78,438,-229,-936,15092,-14996,410,-437,-121,21307
$z_{75}$: 649,766,-429,-459,-82,1427,535,-2991,-174,-1646,511,350,1606,-763,-1441,463,-1352,861,393,86,363,-1351,922,85,719,-17,-167,-263,-20612,-211,-21356
$z_{76}$: -24,-124,-73,88,-67,138,38,129,53,17,-7,279,-23,-49,-350,110,378,51,1,41,-35,170,24,-57,-540,-224,217,21099,21501,-83,128
$z_{77}$: -1,-256,159,192,69,-160,-137,-20,74,230,-351,77,223,-14,-232,-265,211,-310,150,664,-159,959,-1913,105,-757,153,7,-20853,-420,68,21610
$z_{78}$: -31,-386,92,301,97,-41,-370,-66,-58,204,76,72,233,32,-61,-615,348,-139,-153,-26,-474,647,-584,442,-13938,6,15066,-11308,10870,10747,-11107
$z_{79}$: 171,-445,419,286,338,-235,-322,-728,-178,-47,-1042,159,1161,347,-618,-513,-357,-513,255,-594,-68,768,2472,122,29911,31,-199,777,100,-181,971
$z_{80}$: 316,454,-94,-733,96,467,669,-1342,20,-665,-317,301,434,-100,-90,486,-593,843,0,-105,-532,-112,779,-32,-15725,-62,-15033,10463,-10865,-10795,9203
$z_{81}$: 413,-243,291,297,186,-429,-507,331,573,14,-489,173,83,-102,-283,-523,64,64,344,133,89,-173,20,-103,-14197,15044,1,-11127,11026,-10513,-11064
$z_{82}$: 457,-79,332,81,-408,-25,267,132,-891,612,-332,270,82,-514,176,273,-266,-193,-228,27,-277,396,-192,-41,48,-15009,-15121,5,-21247,-60,-540
$z_{83}$: -148,-286,158,-159,0,-223,-79,-309,103,-256,96,50,661,91,-178,-420,-426,-15,59,66,-305,14,-801,143,14565,-28,15029,11179,10343,10455,11275
$z_{84}$: 455,-833,245,237,98,40,453,142,-245,542,302,-479,84,226,111,-143,-201,-15,-671,4,-15,166,-255,-179,-14731,-123,15197,9899,10909,-10581,-11351
$z_{85}$: 187,-131,228,67,28,-105,7,-79,74,64,23,-128,281,99,-88,-185,-46,-70,100,238,-95,-51,26,4,720,176,-261,-21239,5,21357,420
$z_{86}$: 24,-341,308,-24,-40,-9,-31,-663,270,-350,-24,20,1012,351,-330,27,-633,-57,294,-13,7,-20,17,0,14287,-50,-15124,11295,-10751,-10878,10556
$z_{87}$: -579,417,35,-1135,-1063,1246,-683,-39,367,-770,-281,-54,-1555,418,1307,943,388,183,-1659,133,373,-1201,1597,353,812,-63,-35,-375,452,-20869,-21300
$z_{88}$: 230,204,-60,-367,-13,-22,-63,-179,-408,613,-778,-51,10,114,468,28,1157,-929,-497,578,-240,631,-1176,878,71,14900,-15118,282,213,21228,-23
$z_{89}$: 24,563,-608,-235,-115,258,-58,-187,53,-398,82,-40,-472,-86,433,279,-32,610,-98,-43,83,-322,391,58,-721,-15000,15177,68,-620,42,21212
$z_{90}$: -107,-150,60,-348,0,-124,23,-622,-237,-1,236,-96,622,48,1,142,-216,-274,47,-92,402,29,-273,55,-14192,15110,-114,-11113,-10255,10720,-11560
$z_{91}$: -87,-1133,344,2,-441,163,-18,411,826,-1274,-772,394,-717,1996,331,-400,147,130,433,595,495,893,-11,-1373,-26,-15052,-14904,-41,21126,-8,496
$z_{92}$: 87,-109,355,70,-44,154,18,-886,-158,-248,2,193,931,288,-466,-108,-354,-355,-26,-155,-89,89,177,-86,14422,32,15072,11106,-10902,-10864,10508
$z_{93}$: -172,852,-1933,-543,-494,-490,-847,1876,89,-554,-370,86,-1260,-235,564,409,656,1500,-1232,-676,-89,-204,2008,508,911,-117,11,-343,21579,216,-20509
$z_{94}$: -20,-9,79,67,123,-22,-21,-75,-12,44,257,35,174,71,-13,-99,100,-52,54,-274,-34,-131,23,-28,605,177,-243,-21281,-21323,51,-296
$z_{95}$: 86,63,2145,-97,-69,-228,8,108,328,-30,-385,-1492,-16,80,104,99,-159,48,-145,-34,80,69,-582,-66,-1562,-106,244,21296,-409,67,21085
$z_{96}$: 580,234,-616,-623,276,-174,-40,425,-299,-854,-375,-642,4,361,-209,518,-289,26,-79,108,-276,81,-284,782,15675,63,-15169,-10234,-10287,-10542,-10332
$z_{97}$: 77,-179,159,17,-93,232,-115,16,147,86,-46,-32,125,-4,-20,-73,40,-210,-85,52,-83,37,-91,-47,-482,-124,88,21376,71,21234,-194
$z_{98}$: -45,681,-566,-313,21,69,312,-588,170,-681,300,377,-651,66,167,227,-150,873,85,-207,1,-456,436,74,-14838,109,15337,-11088,10458,-10754,10061
$z_{99}$: 71,445,71,584,107,453,22,64,198,-113,254,269,465,-487,-667,-78,-918,252,-158,671,-580,373,534,-582,15793,15119,-283,-10147,-10274,10686,-10160
$z_{100}$: -68,251,-132,-8,-109,-91,5,179,-146,-233,172,82,-227,-182,80,95,281,117,111,-29,-5,182,8,-121,-598,-179,141,21048,55,-21543,-583
$z_{101}$: -47,729,-788,58,207,96,353,-431,143,-570,343,-207,-423,-595,140,210,-182,600,2,-303,-76,-463,331,126,-14967,-15237,60,-10997,10497,10605,10255
$z_{102}$: 268,841,-655,-129,-739,-182,584,126,519,-580,-473,323,-2111,-783,727,1002,133,1072,73,-86,-238,-350,1275,-396,914,14998,14903,-215,479,237,-21150
$z_{103}$: -61,632,-391,-263,32,-203,179,-61,-238,-245,86,-151,-541,-333,295,609,-155,750,-261,87,50,-601,792,-153,-15321,-6,-14899,-10946,10518,-10681,10184

$z_{104}$: -131,288,212,-5,59,300,237,891,-5,360,-240,-209,-835,-115,209,45,146,115,-234,20,61,218,212,-74,14508,-15168,49,11155,-10818,10428,10741
$z_{105}$: -48,166,44,421,300,-261,198,184,69,288,550,500,-747,9,-646,66,215,-387,224,-907,-860,-1294,2493,-326,1548,61,-197,-21417,452,158,-20860
$z_{106}$: -49,0,-218,-21,18,-93,-17,136,-8,24,21,-287,14,66,-88,-106,103,-71,16,18,-34,-37,-77,1,-488,-78,31,21342,-21255,-56,-814
$z_{107}$: -28,300,-552,439,604,-1695,-281,-916,180,-1819,-326,-1499,2816,-36,-305,-414,-1363,-10,-166,234,348,-182,1101,1207,-1009,57,78,175,20681,85,21360
$z_{108}$: 448,156,131,405,787,-123,-169,-1181,27,293,-91,-90,434,42,16,273,14,8,-185,-493,-1188,97,383,-470,15153,-15080,50,10783,10864,10628,-9960
$z_{109}$: -788,-620,625,1447,825,-1065,1576,-786,-537,108,-311,616,2214,321,-588,-344,-2540,-1350,474,715,-1223,2142,-1741,-473,62,65,-132,-69,-20578,-21228,-1128
$z_{110}$: -265,291,-492,-225,-123,58,130,-680,-51,-780,194,73,-438,-285,211,76,-352,782,230,-203,38,-499,687,-3,-15041,15257,37,-10844,10518,10674,10186
$z_{111}$: -196,74,-216,2,541,-72,-366,-321,86,-829,-93,741,359,-515,-303,279,-145,-267,452,-385,269,367,150,279,15234,1,14933,10871,-10411,10669,-10475
$z_{112}$: 0,-159,61,65,84,-101,-86,-24,-90,374,-19,-45,-268,-287,-3,129,144,-104,-154,86,73,-8,-43,-84,647,147,-134,-21213,45,-21388,42
$z_{113}$: -692,837,-656,-601,578,36,214,-391,-98,-665,247,-172,-405,47,321,497,-309,618,125,-246,-529,-353,360,12,-15732,-145,-14918,10441,10625,10664,9903
$z_{114}$: 38,-139,325,-3,-105,343,53,-638,-3,-18,489,-8,736,84,-134,-352,156,-84,-64,42,-129,160,-325,-61,-14791,-77,15123,9981,-10395,10664,-11735
$z_{115}$: 338,948,362,990,-1620,-7,76,-876,743,-19,-403,467,1047,-590,-269,-505,54,-396,1092,1391,-536,1460,-688,-102,-29,-83,-29897,222,234,-108,-158
$z_{116}$: 167,-114,375,33,-110,41,47,562,-165,705,-59,-435,-242,-298,271,-165,232,-228,126,152,19,359,-260,322,15107,103,14965,-10200,10238,-10681,11459
$z_{117}$: -295,-13,381,446,-124,479,-263,9,-264,-52,184,57,880,456,708,314,-530,16,443,-261,142,-454,-45,-188,15131,-15057,-12,10769,-10350,-10601,-10677
$z_{118}$: -91,-57,270,-3,124,125,-47,-241,-107,-11,106,79,-20,-47,-55,-196,168,-330,-8,-42,76,54,-379,19,601,15255,-15320,-20977,133,2,158
$z_{119}$: 112,572,-544,-184,61,-199,394,-309,-67,-706,-193,-577,-363,116,514,543,-675,808,-344,-60,149,-243,918,-76,-15367,-88,15532,10195,10581,10788,9695
$z_{120}$: -379,-285,-3046,-250,-343,388,-97,1214,-505,525,-818,1837,-688,115,6,49,371,-360,170,808,-37,811,-1342,97,585,-220,45,20908,679,106,-21227
$z_{121}$: -383,243,-480,-209,-177,799,174,-80,-231,-247,-153,326,-277,93,-8,-154,166,478,49,-392,-15,181,-127,411,14694,56,-15216,-9917,10506,-10823,11529
$z_{122}$: 23,667,-582,-322,147,453,506,103,0,151,170,95,-1238,-122,716,766,75,352,-335,-136,-213,-118,458,-169,-14988,129,15127,-11032,-10868,10831,9652
$z_{123}$: -530,-462,254,875,252,-21,633,-3831,-713,-1455,-437,-673,2910,522,-1673,425,-3136,236,125,621,693,-890,428,346,792,-75,6,-142,965,20399,-21151
$z_{124}$: -351,-429,134,592,-133,-818,98,1747,-712,49,-989,-830,-76,-265,146,80,-90,-425,-182,508,-746,1013,-371,-402,47,15026,15057,-59,1,-21136,-242
$z_{125}$: -168,229,-192,-193,137,127,-93,144,-86,-3,93,0,-43,-26,259,-48,257,-5,-57,-66,-5,-390,180,185,-936,-15098,-14979,263,-465,-53,21314
$z_{126}$: 0,-1,0,0,0,0,0,0,-1,0,10675,0,0,0,0,0,-1,0,0,10637,0,10655,-10638,0,-535,98,-101,-21273,-133,31,-1313
$z_{127}$: 0,0,0,0,0,0,0,0,0,0,10641,0,0,0,0,0,0,0,0,10708,0,-10669,10588,0,-509,21,-223,-21241,-33,-153,1749
$z_{128}$: 0,0,0,0,0,0,0,0,0,0,10668,0,0,0,0,0,0,0,0,-10639,0,10674,10625,0,-535,-35,-97,-21268,123,2,1386
$z_{129}$: 0,0,0,0,0,0,0,0,0,0,10652,0,0,0,0,0,0,0,0,-10645,0,-10653,-10655,0,-519,-26,-149,-21311,288,90,240
$z_{130}$: 0,0,0,0,0,0,0,0,0,0,-10654,0,0,0,0,0,0,0,0,10647,0,10653,10651,0,-522,293,-183,-21308,-352,-28,200
$z_{131}$: 0,0,0,0,0,0,0,0,0,0,-10652,0,0,0,0,0,0,0,0,10651,0,-10652,-10651,0,-548,173,-215,-21286,-250,3,-1021
$z_{132}$: 0,0,0,0,0,0,0,0,0,0,-10650,0,0,0,0,0,0,0,0,-10680,0,10658,-10618,0,-504,59,-68,-21286,-107,294,-1048
$z_{133}$: 0,0,0,0,0,0,0,0,0,0,-10722,0,0,0,0,0,0,0,0,-10675,0,-10616,10593,0,-485,183,-110,-21211,-23,31,2085
$z_{134}$: -1,0,0,1,-1,2,0,-2,1,0,10648,1,1,1,0,-1,1,1,-1,10695,0,10652,-10607,-2,14264,9,-34,6358,-89,39,14517
$z_{135}$: 0,0,0,0,0,0,0,0,0,0,10671,0,0,0,0,1,0,0,-1,10646,0,-10658,10631,0,12817,-15,7,7259,9,-16,15415
$z_{136}$: 0,0,0,0,0,0,0,0,0,0,10673,0,0,0,0,0,0,0,0,-10670,0,10666,10597,0,12910,-13,6,7151,4,-10,15388
$z_{137}$: 0,0,0,0,0,0,0,0,0,0,10669,0,0,0,0,0,1,0,0,-10624,0,-10661,-10643,0,14276,5,-16,6347,-7,1,14514
$z_{138}$: 0,0,0,0,0,0,0,0,0,0,-10651,0,0,0,0,1,0,0,-1,10637,0,10652,10666,0,13034,-16,-7,6858,6,-3,15416
$z_{139}$: 0,0,0,0,0,0,0,0,0,0,-10664,0,0,0,0,0,-1,1,0,10686,0,-10668,-10587,-1,14287,24,4,6393,-7,14,14478
$z_{140}$: 0,0,0,0,0,0,0,0,0,0,-10657,0,0,0,0,0,0,0,0,-10633,0,10651,-10655,0,14104,8,1,6545,3,-18,14596
$z_{141}$: 0,0,0,0,0,0,0,0,0,0,-10639,0,0,0,0,0,0,0,0,-10673,1,-10672,10622,0,13023,-10,2,6885,7,2,15413
$z_{142}$: 0,0,0,0,0,0,0,0,0,0,10651,0,0,0,0,0,0,0,0,10673,0,10651,-10630,0,-14303,0,0,6300,67,-67,-14501
$z_{143}$: 1,-2,2,0,1,-1,-1,-1,-1,0,10653,2,-2,0,0,1,-1,0,1,10682,-1,-10647,10622,-1,-13074,34,54,7600,-20,55,-15031
$z_{144}$: 0,0,0,0,0,0,0,0,0,0,10664,0,0,0,0,0,0,0,0,-10633,0,10668,10641,0,-12988,44,37,6934,9,11,-15421
$z_{145}$: 0,0,0,0,0,0,0,0,0,0,10652,0,0,0,0,0,0,0,0,-10638,0,-10659,-10658,0,-13197,-40,-8,6501,18,-13,-15431
$z_{146}$: 0,0,0,0,0,0,0,0,0,0,-10684,0,0,0,0,0,0,0,0,10614,0,10619,10689,0,-14235,157,-91,7204,-35,93,-14141
$z_{147}$: 0,0,0,0,0,0,0,0,0,0,-10665,0,0,0,0,0,0,0,0,10643,0,-10664,-10634,0,-14293,1,1,6271,17,-17,-14524
$z_{148}$: 0,0,2,0,3,-2,-1,1,2,-1,-10648,-2,1,1,0,3,0,1,1,-10692,0,10650,-10611,-1,-14290,-16,1,6264,-13,-10,-14533
$z_{149}$: 0,0,0,0,0,0,0,0,0,0,-10652,0,0,0,0,0,0,0,0,-10652,0,-10652,10652,0,-12622,288,157,7500,-475,519,-15439
$z_{150}$: 0,0,10598,0,0,10705,0,0,10705,0,0,-10598,0,0,0,0,0,0,0,0,0,0,0,0,521,69,-96,21239,51,-331,-1755
$z_{151}$: 0,0,10656,0,0,10649,0,0,-10647,0,0,10655,0,0,0,0,0,0,0,0,0,0,0,0,517,-186,18,21310,-234,-223,-185
$z_{152}$: 0,0,10657,0,0,-10653,0,0,10646,0,0,10650,0,0,0,0,0,0,0,0,0,0,0,0,-531,-12,110,21293,-131,-97,-913
$z_{153}$: 0,0,10604,0,0,-10681,0,0,-10699,0,0,-10622,0,0,0,0,0,0,0,0,0,0,0,0,520,-129,169,21247,165,10,-1667
$z_{154}$: 0,0,-10575,0,0,10661,0,0,10729,0,0,10641,0,0,0,0,0,0,0,0,0,0,0,0,499,-52,-159,21227,-137,-192,1907
$z_{155}$: 0,0,-10637,0,0,10637,0,0,-10666,0,0,-10666,0,0,0,0,0,0,0,0,0,0,0,0,514,-322,174,21299,206,-84,667
$z_{156}$: 0,0,-10641,0,0,-10661,0,0,10662,0,0,-10642,0,0,0,0,0,0,0,0,0,0,0,0,509,-49,102,21311,326,39,-75
$z_{157}$: 0,0,-10607,0,0,-10679,0,0,-10696,0,0,10624,0,0,0,0,0,0,0,0,0,0,0,0,559,-157,194,21213,54,273,2027
$z_{158}$: 0,1,1,1,0,0,0,10658,10644,0,0,0,0,10646,-10656,0,0,0,0,0,1,0,0,1,10926,-1,10161,-10899,-7496,7551,-27
$z_{159}$: 1,-1,0,1,-1,1,1,10671,10630,-1,0,0,0,-10631,10672,0,-1,-2,0,1,0,-1,0,0,10395,0,10408,-10408,-8058,8059,0

$z_{160}$: 0,0,-1,0,-1,0,0,10652,-10651,0,0,0,0,10651,10651,1,0,0,1,0,0,0,0,0,10836,20,10846,-10946,-7454,6635,366
$z_{161}$: 0,0,2,-2,-2,0,0,10651,-10651,-1,2,1,0,-10651,-10650,-2,-1,-2,2,0,-2,-2,1,2,10373,-1,11295,-10421,-7427,7463,-18
$z_{162}$: 1,-5,-5,1,-4,-3,-2,-10646,10652,-2,-4,2,-2,10651,10648,1,0,1,0,4,0,-1,4,-1,10380,-1,11311,-10415,-7416,7455,-19
$z_{163}$: 0,-1,-1,-1,0,0,2,-10652,10651,0,-1,0,0,-10650,-10651,1,1,1,0,0,0,0,-1,1,10423,-10,10465,-10358,-8038,8034,8
$z_{164}$: -1,1,2,2,0,2,2,-10636,-10667,0,0,1,0,10667,-10632,-1,1,-2,0,1,0,0,0,0,10402,19,10401,-10398,-8065,8070,11
$z_{165}$: 0,-1,0,0,1,-1,0,-10651,-10652,0,1,0,-1,-10651,10649,0,0,-1,2,1,-2,0,1,0,11193,312,10480,-10524,-7350,7389,197
$z_{166}$: 399,-5060,5354,1764,-12,-2349,-2660,1162,806,3881,-232,609,4549,1208,-2527,-4375,997,-5754,885,1524,-276,3817,
-4836,785,-21237,-38,415,-98,-243,-71,15684
$z_{167}$: -267,-1455,535,-307,-353,-1814,-117,-8811,543,-5524,-3372,180,9759,392,-1017,-484,-721,-59,1016,462,-369,-577,
-1191,546,-799,114,-453,472,-16163,14868,14091

## Use of artificial intelligence

GPT-6-Astra Pro and Claude Opus 5.5 were used to find these constructions and draft this paper.

## Data availability

The coordinates and an exact checking program accompany this paper.

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